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SolveBench

Direct and Iterative Linear Solvers on Real Sparse Matrices: A Diagonal-Selection Pipeline for Stationary Iterative Methods

Course Project Report
CSE 402: Numerical Analysis Lab

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1 Introduction

This report describes SolveBench, a course project built for CSE 402 that takes a small, elegant piece of published research on Jacobi and Gauss-Seidel iteration and asks what happens to its conclusions once real engineering matrices, rather than hand-picked or randomly generated ones, are put in front of it. What began as a verification exercise turned into something with more moving parts than we expected: a 927-matrix benchmark across sixteen solver families, a discovery that over half of that corpus cannot even be handed to the base paper’s methods, a preprocessing pipeline built in response to that discovery, and, in the final two weeks of the project, a self-audit that overturned one of our own headline claims and replaced it with a smaller, better-supported one. This introduction lays out the problem, the scope of what we built, and how the rest of the report is organized.

1.1 The problem: solving $A\vec{x} = \vec{b}$ at scale

Almost every simulation of a physical system, a power grid under load, a bridge under stress, an integrated circuit under bias, a chemical plant at steady state, eventually reduces to one piece of linear algebra:

$$A\vec{x} = \vec{b}$$

Here A encodes how the parts of the system interact with each other, $\vec{b}$ is whatever is being pushed into the system, and $\vec{x}$ is the quantity being solved for. For a problem of any realistic size, A has thousands, sometimes millions, of rows and columns, but it is almost always *sparse*: the overwhelming majority of its entries are exactly zero, because most components of a physical system only interact with their immediate neighbours. A resistor in a circuit is wired to the two or three nodes touching it, not to every other node in the network.

Two broad families of technique exist for solving such a system. **Direct methods** factor A into triangular pieces, in the same spirit as Gaussian elimination done by hand, and then solve two easy triangular systems. This produces the exact answer, up to floating-point rounding, in a fixed and predictable number of steps, but the factorization step can introduce a great many new nonzero entries where A had zeros (*fill-in*), and for a sufficiently large sparse matrix this can exceed the memory available. **Iterative methods** instead start from a guess, usually $\vec{x}_0 = \vec{0}$, and repeatedly refine it, checking the residual $\|A\vec{x} - \vec{b}\|$ at every step and stopping once it is small enough. Each step is comparatively cheap, essentially one matrix-vector product, so iterative methods scale to far larger problems than direct factorization can reach. The catch is that nothing guarantees the steps actually approach the answer; whether they do depends entirely on the structure of A , and that dependency is the subject of this entire project.

1.2 Problem statement

The concrete question this project set out to answer is simple to state and, as it turned out, considerably harder to answer honestly than we expected at the outset:

- A well-known paper (Section 2.1) derives, by hand, exactly which small matrices make Jacobi and Gauss-Seidel converge, and validates that theory statistically on randomly generated matrices with at most five unknowns. **Does any of that survive contact with the matrices that working engineers actually have to solve, at the sizes those matrices actually come in?**

- Where the classical iterative methods do fail, is the failure a property of the *matrix*, in which case nothing can be done about it short of switching solver families, or is it a property of an arbitrary, fixable choice made somewhere upstream of the solver, such as the order in which the matrix's rows happen to be listed?
- If some of that failure is fixable, how much of it, by what mechanism, and at what cost? A preprocessing step that helps on paper but costs more than it saves, or that quietly breaks some other case while fixing this one, is not a contribution; it is a trade that needs to be measured, not assumed.

1.3 Scope and objectives

As actually built and measured, the project delivers the following, each of which is described in full later in this report:

- A from-scratch implementation of Jacobi, Gauss-Seidel, and Successive Over-Relaxation (SOR), together with the classical convergence theory that governs them, built independently of any existing solver library (Section 4).
- A benchmark harness that runs sixteen solver methods, spanning direct, stationary iterative, Krylov, and preconditioned families, across every matrix in a 927-matrix corpus drawn from the SuiteSparse Matrix Collection, with success decided independently from the measured residual rather than from any solver's own claim (Section 3, Section 6).
- A diagnosis of why the classical stationary methods are unusable on more than half of that corpus before a single iteration is attempted, and a proof, not merely a guess, of which preprocessing operations could possibly change that (Section 5).
- A preprocessing pipeline, built on that proof, that chooses which row of the matrix sits on the diagonal and, for SOR, chooses its relaxation factor per matrix, and a full measurement of what that pipeline does and does not buy (Section 5, Section 6).
- A dedicated audit of our own results, carried out in the final phase of the project, that retracted one headline claim, explained precisely why a second ambitious idea underperformed its own theoretical promise, and produced a cleaner, better evidenced finding in its place (Section 7).

The project deliberately targets square, real-valued sparse matrices with at most ten thousand rows and columns; it does not attempt multigrid or domain-decomposition methods, GPU or distributed execution, or wall-clock performance engineering, and it does not propose a new iterative method. Every one of these boundaries is a considered decision, not an oversight, and each is revisited where it matters in the discussion (Section 8).

1.4 How this report is organized

Section 2 introduces the base paper this project extends and places it against the small body of related work on diagonal reordering for iterative solvers. Section 3 describes the SuiteSparse-derived corpus in detail, with concrete named examples. Section 4 covers every method implemented directly from the base paper, together with the classical convergence theory and the additional solver families built so the comparison would have a floor and a ceiling. Section 5 tells the story of the project's own contribution as it actually happened, milestone by milestone, including the false starts. Section 6 reports every measured number behind the headline claims. Section 7 is a dedicated failure analysis, requested specifically for this report, that explains why each solver family fails on this corpus and how both the classical methods and our own designed methods were investigated and, where possible, improved in response. Section 8 discusses honest strengths and limitations and points to future work, and Section 9 closes the report.

2 Motivation and Related Work

2.1 The base paper: Khrapov and Volkov, 2024

This project extends a specific, published piece of research: P. Khrapov and N. Volkov, *Comparative Analysis of Jacobi and Gauss-Seidel Iterative Methods*, International Journal of Open Information Technologies, vol. 12, no. 2, pp. 23–34, 2024. The paper studies the two oldest and simplest stationary iterative methods for $A\vec{x} = \vec{b}$. For systems of two and three unknowns, the authors derive, algebraically and by hand, the precise region of coefficient space in which each method converges and in which it diverges. They then generate a large number of random matrices with up to five unknowns and check, empirically, how often the derived theory holds.

What the paper does not do is, in a sense, more informative than what it does. It never once evaluates a matrix that came from an actual engineering problem. Every matrix studied is either small enough to be solved by hand or generated by a random-number routine, and a randomly generated matrix does not resemble a real one in any structural sense. A matrix that comes from a circuit simulator, a finite-element mesh, or a power-flow model carries a specific, non-random pattern of zero and nonzero entries determined by which physical components are actually connected to which others. This is precisely the gap this project set out to close: nobody, as far as we could determine, had checked whether the base paper’s conclusions survive contact with matrices that engineers actually have to solve, at the sizes those matrices actually arrive in.

2.2 Related work on diagonal selection for iterative solvers

The idea of choosing which entry of a sparse matrix sits on the diagonal, rather than accepting whatever order the matrix arrived in, is not new; it is a well established tool in the numerical linear algebra literature, though almost entirely in service of a different goal than the one this project pursues.

Duff and Koster’s algorithm for permuting large entries onto the diagonal, published as *The design and use of algorithms for permuting large entries to the diagonal of sparse matrices* (SIAM Journal on Matrix Analysis and Applications, vol. 20, pp. 889–901, 1999) and distributed widely as the routine MC64, is the standard tool this project measures its own idea against throughout. MC64 chooses the diagonal that maximizes the product of the diagonal magnitudes, a criterion built specifically to make subsequent direct factorization numerically stable. Hook, Pestana, Tisseur, and Hogg extended this line of work in *Max-Balanced Hungarian Scalings* (SIAM Journal on Matrix Analysis and Applications, vol. 40, no. 1, pp. 320–346, 2019), addressing the fact that a maximum-product assignment is often not unique and proposing a principled way to choose among the ties from the point of view of diagonal dominance, a concern that turns out to be close in spirit, though not in method, to the objective this project builds in Section 5.6. Benzi, Haws, and Tuma, in *Preconditioning highly indefinite and nonsymmetric matrices* (SIAM Journal on Scientific Computing, vol. 22, pp. 1333–1353, 2000), studied permutation and scaling strategies aimed specifically at making incomplete factorizations more reliable on hard, indefinite systems, a concern that resurfaces directly in this project’s own encounter with incomplete LU preconditioning (Section 7.7). None of this prior work, to our knowledge, poses the question this project asks: can a row permutation be chosen specifically to make the classical stationary methods, Jacobi, Gauss-Seidel, and SOR, converge, rather than to stabilize a direct factorization or a preconditioner. That distinction, and everything it implies once measured honestly, is the actual substance of Section 5.

Barrett et al.’s *Templates for the Solution of Linear Systems* (SIAM, 1994) is the standard reference for the broader family of Krylov and preconditioned methods implemented in this project purely as context, described in Section 4.7, and it is also, as it happens, the source of a lesson the group learned the hard way about its own original proposal, told in full in Section 5.1.

3 Datasets

3.1 The SuiteSparse Matrix Collection

The SuiteSparse Matrix Collection (Davis and Hu, *The University of Florida Sparse Matrix Collection*, ACM Transactions on Mathematical Software, vol. 38, no. 1, 2011) is the standard public library of real, sparse matrices contributed by engineers and scientists from actual projects, each tagged with metadata describing its size, its number of nonzero entries, and the real-world domain it came from. This is what lets the project ask a question the base paper’s random matrices cannot: does Gauss-Seidel actually work on a circuit-simulation matrix, rather than does it work on numbers chosen uniformly at random.

3.2 Selection rule and corpus bookkeeping

The selection rule is every square, real-valued matrix in the collection with at most ten thousand rows and columns. That cap is a memory limit, not a judgement about difficulty: one of the project’s analyses computes the exact spectral radius of the iteration matrix, which requires a dense $n \times n$ array, and at $n = 10,000$ that array alone already occupies roughly 800 megabytes. The collection holds 1,184 matrices meeting this size criterion; this project draws a sample of 830 of them, not a complete census, together with 100 matrices generated by the project itself specifically to reproduce the base paper’s own validation approach at larger scale, for a nominal total of 930 downloaded files.

That nominal total needs three corrections before any result can be trusted. Three files are exact byte-for-byte duplicates of other files under a different name (`bcsstk07`≡`bcsstk06`, `bcsstk12`≡`bcsstk11`, `t2da1_a`≡`t2da1`), bringing the unique corpus to **927 matrices**. Six further files turned out to be truncated downloads, each one’s own header declaring more nonzero entries than the file physically contained, so they fail to load at all (`cavity20`, `nemeth17`, `nemeth22`, `nemeth23`, `nemeth25`, `psmigr_2`); Section 7.4 explains how this was caught. Finally, one physical domain, computational fluid dynamics, arrived split across two folders because SuiteSparse’s own metadata spells it two different ways depending on which research group contributed the matrix; the two are merged for every table and figure in this report. None of these corrections change any conclusion reported later; they only change which denominator a rate is computed against, and every table in this report states that denominator explicitly rather than leaving it implicit.

3.3 The 26 domains

Table 1 lists every domain in the corpus with its matrix count. The distribution is sharply uneven: one domain alone, circuit simulation, contributes over a fifth of the entire corpus, while eight domains contribute fewer than five matrices each. This is not a defect in the corpus; it is a fact about which of these problems practitioners happen to solve most often, and it is the reason every per-domain claim made anywhere in this report states its own denominator rather than leaving the reader to assume a comparable sample size across domains.

Table 1: The 26 real-world domains represented in the corpus, by matrix count.

Domain	Matrices
Circuit simulation	200
Computational fluid dynamics	121
Random matrices (self-generated)	100
Structural engineering	87
Optimal control	78
2D/3D discretization	49
Chemical process simulation	44
Optimization	39
Undirected weighted graph	37
Theoretical quantum chemistry	27
Electromagnetics	25
Eigenvalue / model reduction	19
Directed weighted graph	17
Power network	17
Economic modelling	16
Model reduction	13
Materials	9
Statistical / mathematical	6
Counter-example	4
Semiconductor device	4
Acoustics	2
Thermal	2
Computational chemistry	1
Computer graphics / vision	1
Linear programming	1
Robotics	1

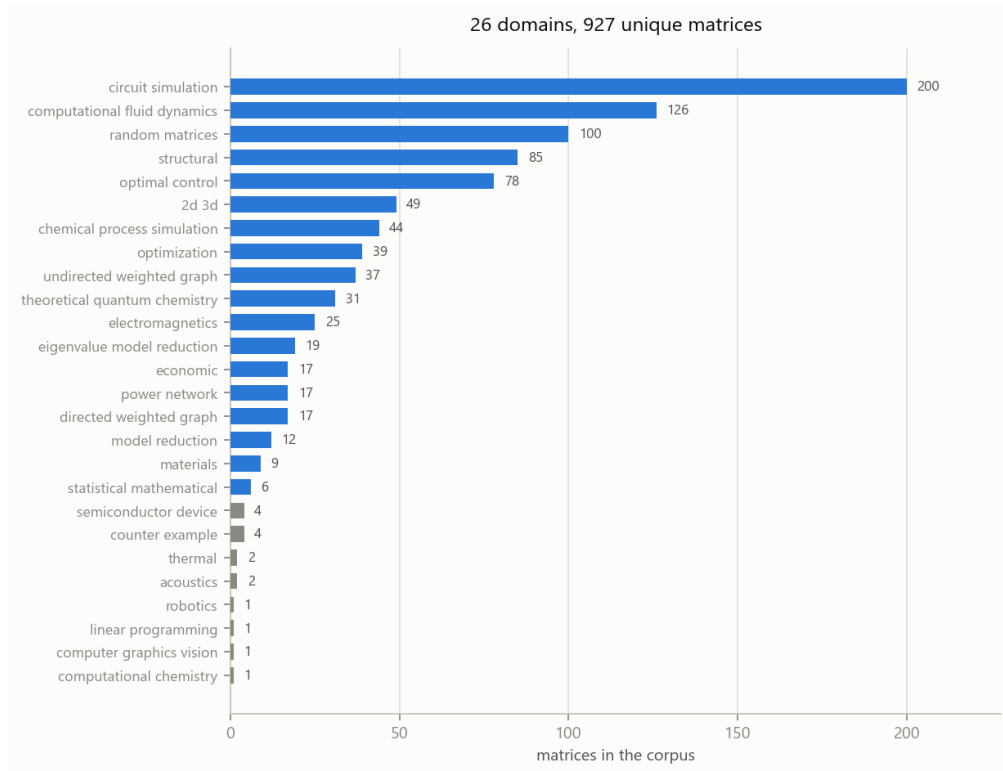


Figure 1: Domain composition of the 927-matrix corpus, drawn to scale. Circuit simulation and computational fluid dynamics together account for close to a third of the entire corpus, while more than half of the 26 domains have fewer than 25 matrices apiece.

3.4 Concrete examples

Table 2 names specific matrices from the corpus so that later claims in this report can be traced back to an actual, inspectable system rather than staying abstract.

Table 2: Representative matrices from the corpus, by domain, size, and nonzero count.

Domain	Matrix	Size (n)	Nonzeros
Circuit simulation	meg1	2,904	58,142
Circuit simulation	circuit_1	2,624	35,823
Structural engineering	Kuu	7,102	173,651
Structural engineering	raefsky5	6,316	168,658
Fluid dynamics	ex40	7,740	458,012
Fluid dynamics	GT01R	7,980	430,909
Power network	TSC_OPF_1047	8,140	1,012,521
Power network	TSC_OPF_300	9,774	415,289
Quantum chemistry	nemeth26	9,506	760,633
Economic modelling	psmigr_3	3,140	543,162
Electromagnetics	fp	7,548	848,553
Self-generated random	rand_100_n300	300	90,000

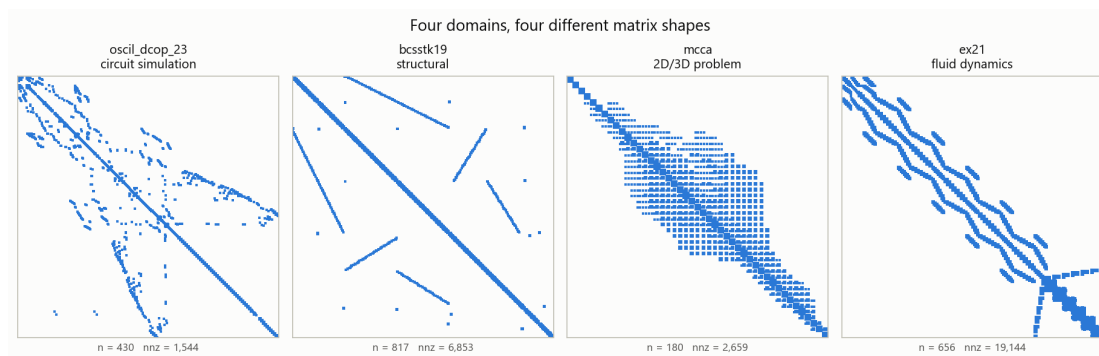


Figure 2: Sparsity patterns of a gallery of corpus matrices spanning several domains. Each panel is the same square canvas at the same visual scale; the density and banding differences between, say, a circuit-simulation matrix and a structural one are a direct, visible reason the two behave so differently under an iterative solver.

A handful of matrices recur throughout this report by name because each one is the concrete evidence behind a specific claim, not an anecdote. `cade6` (fluid dynamics, $n = 961$) has a Jacobi spectral radius of 0.717, comfortably inside the convergent region, yet its residual grows to 44,764 times its starting value before it finally shrinks and converges at iteration 178; this single matrix is the reason the harness’s divergence guard has to be lenient, a point returned to in Section 5.2. `d_ss` (chemical process simulation, $n = 53$) is the one matrix in the whole corpus on which the project’s own reordering objective makes the spectral radius worse, from 1.884 to 29.605, discussed honestly in Section 7.9. `nnc261`, `west0067`, `bcsstk19`, and `rw5151`, four matrices from four different domains, each separately exposed a hang in an off-the-shelf scipy matching routine, the subject of Section 5.7. `iprob` (linear programming, $n = 3,001$) is one of only four matrices in the entire corpus that the project’s own bottleneck objective rescues and the established MC64 objective does not, discussed in Section 5.8.

4 Methods Implemented From the Base Paper

Every method the base paper studies was implemented independently, from first principles, before any comparison against real data was attempted. This section presents each one together with the convergence theory that governs it, and then describes the further solver families added purely so the comparison would have context: a specialised method looking good or bad in isolation means little without a floor and a ceiling to measure it against.

4.1 Splitting the matrix

Jacobi, Gauss-Seidel, and SOR all begin from the same decomposition of the coefficient matrix into three pieces:

$$A = D + L + U$$

where D holds only the diagonal entries, L is everything strictly below the diagonal, and U is everything strictly above it. Every one of the three methods needs to divide by the entries of D at some point in its update rule, a fact whose consequences turn out to be far larger than a technical footnote, as Section 5.3 describes.

4.2 Jacobi's method

Jacobi's update uses only values from the previous complete iteration; nothing computed so far in the current pass feeds back into itself.

```
function JACOBI(A, b, tolerance, max_iterations):
    x <- zeros(n)
    for k in 1 .. max_iterations:
        r <- b - A @ x
        if norm(r) / norm(b) <= tolerance:
            return x, converged = true
        x <- x + r / diagonal(A)
    return x, converged = false
```

4.3 Gauss-Seidel

Gauss-Seidel is the same idea with one change: each row's update can immediately use the already-updated values from earlier rows in the same pass. This usually converges noticeably faster than Jacobi on the same matrix, though not always, as Section 6.3 measures directly.

```
function GAUSS_SEIDEL(A, b, tolerance, max_iterations):
  x <- zeros(n)
  for k in 1 .. max_iterations:
    r <- b - A @ x
    if norm(r) / norm(b) <= tolerance:
      return x, converged = true
    solve (D + L) @ delta = r    for delta
    x <- x + delta
  return x, converged = false
```

4.4 Successive Over-Relaxation (SOR)

SOR scales the Gauss-Seidel update by a relaxation factor ω before applying it, deliberately overshooting the correction, which can converge faster than plain Gauss-Seidel when ω is chosen well. Setting $\omega = 1$ reduces exactly to Gauss-Seidel.

```
function SOR(A, b, omega, tolerance, max_iterations):
  x <- zeros(n)
  for k in 1 .. max_iterations:
    r <- b - A @ x
    if norm(r) / norm(b) <= tolerance:
      return x, converged = true
    solve (D + omega * L) @ delta = omega * r    for delta
    x <- x + delta
  return x, converged = false
```

SOR was not itself the subject of the base paper, but it is Gauss-Seidel with a single extra tuning parameter, and that parameter turns out to matter a great deal to this project's own contribution, described starting in Section 5.10.

4.5 Convergence theory: the spectral radius

Whether any of the three methods above converges at all is governed by a single number: the spectral radius $\rho(T)$ of that method's iteration matrix T (a different T for Jacobi, $T_J = -D^{-1}(L + U)$, than for Gauss-Seidel, $T_{GS} = -(D + L)^{-1}U$).

If $\rho(T) < 1$: the error shrinks a little further every step, eventually to zero.

If $\rho(T) \geq 1$: the error does not shrink; the method will not converge.

The spectral radius also governs speed, not merely whether convergence happens at all: reaching a tolerance ε takes roughly $\log(\varepsilon)/\log(\rho(T))$ steps. A ρ of 0.5 needs about 27 steps to reach 10^{-8} ; a ρ of 0.999 needs roughly 18,000, and, as Section 7.2 shows, this gap between “converges” and “converges usefully” turns out to be the single most important fact about this entire corpus.

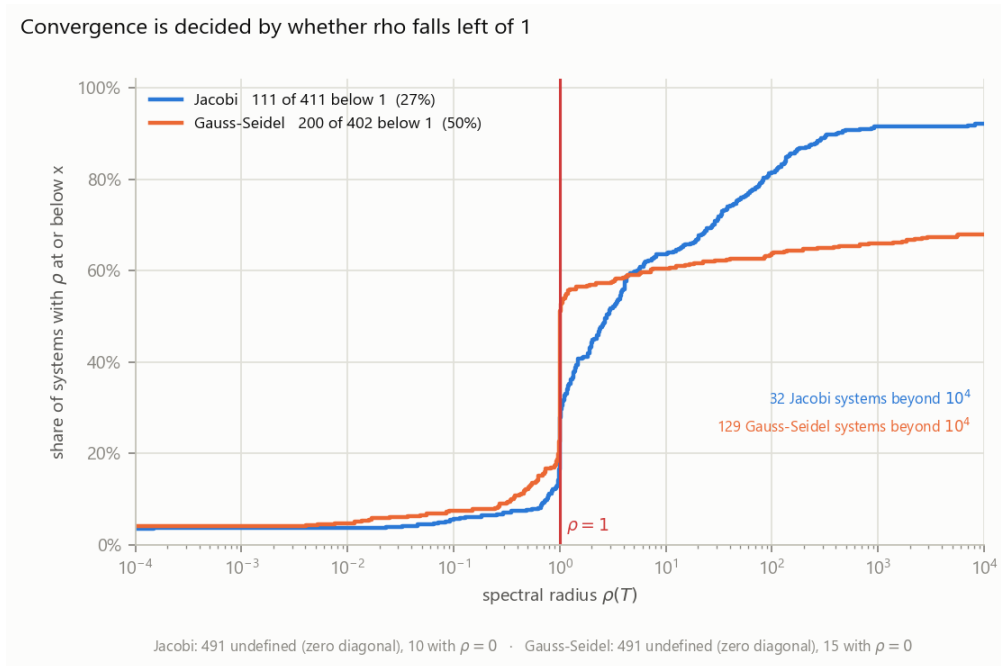


Figure 3: Distribution of $\rho(T_{Jacobi})$ and $\rho(T_{Gauss-Seidel})$ across the corpus. The mass sitting just above and just below 1 is not a rounding artefact; it is the regime that decides most of what Section 7 has to explain.

Beyond the spectral radius itself, a family of classical sufficient conditions, strict and weak diagonal dominance, the H-matrix, M-matrix, and L-matrix properties, symmetric positive definiteness, and the property-A structure behind the classical Householder-John theorem, each guarantee convergence without requiring $\rho(T)$ to be computed directly. Figure 4 shows how much of the corpus each classical hypothesis actually covers; Section 7.2 shows what happens to the matrices none of them cover.

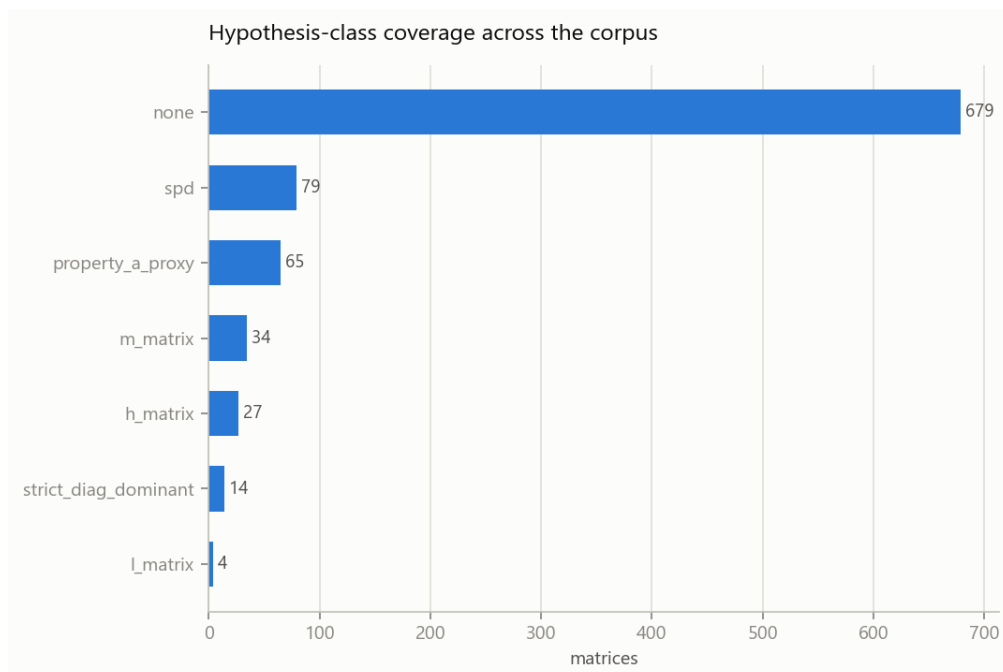


Figure 4: How many corpus matrices fall under each classical convergence-guarantee hypothesis. The classes overlap (a strictly diagonally dominant matrix is also an H-matrix, for instance), and a large fraction of the corpus falls under none of them.

4.6 Reproducing the base paper’s own experiment

To compare against the base paper on its own terms, not only on new terms of our choosing, the project generated its own `random_matrices` domain, 100 matrices sized from 5 to 300 unknowns, following the same generation procedure as the base paper’s own validation experiment. This is a deliberate methodological choice: it lets Section 6.3 report how the base paper’s own test bed behaves once matrices larger than five unknowns are allowed, rather than only reporting how our new, larger corpus behaves, which would leave the comparison one step removed from the paper it is meant to be checking.

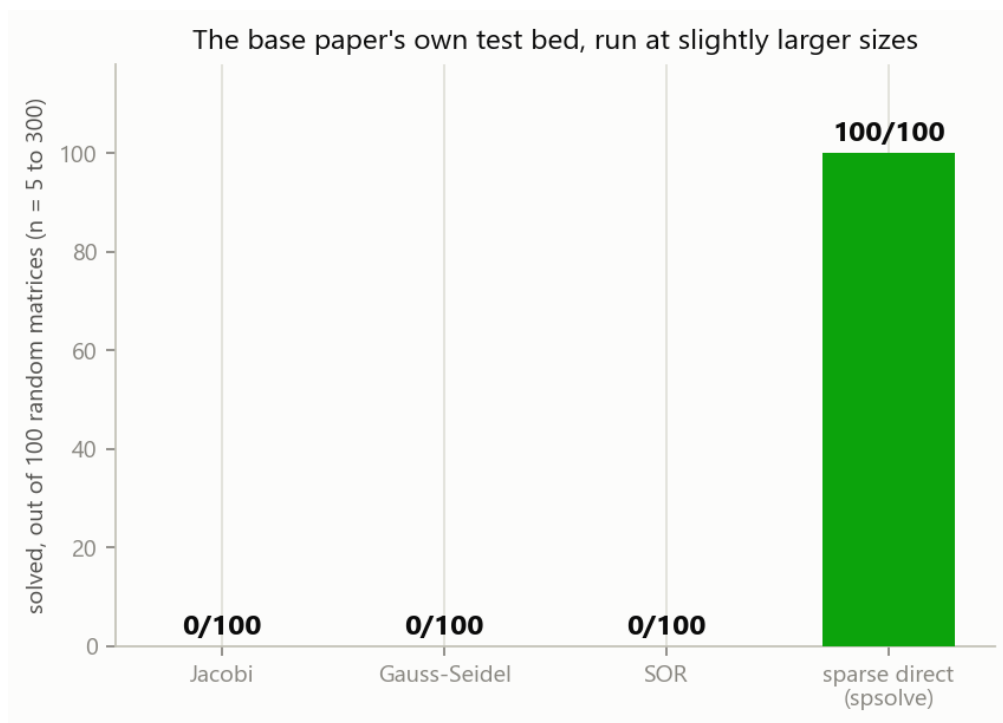


Figure 5: Spectral radius against matrix size for the self-generated random-matrix domain. The growth is visible almost immediately past the base paper’s own tested range of two to five unknowns.

4.7 Additional baseline families

The base paper studies exactly two methods. Reporting that Jacobi or Gauss-Seidel succeeds or fails on some fraction of a corpus means little without something to compare that fraction against, so thirteen further methods across three additional families were implemented, giving sixteen methods in total.

4.7.1 Direct methods

Gauss elimination, Gauss-Jordan elimination, LU decomposition, and Cholesky decomposition were implemented by hand, in addition to the industrial-strength sparse routines `spsolve` and `splu` from the SuperLU library accessed through `scipy`. Direct methods factor the matrix once and then solve two triangular systems, giving the exact answer, up to floating-point rounding, in a fixed number of steps, at the cost of potentially large memory use from fill-in.

4.7.2 Krylov methods

Conjugate Gradient, BiCGSTAB, and GMRES(30) build up a small search space by repeatedly multiplying the matrix against a vector, $\vec{b}$, $A\vec{b}$, $A^2\vec{b}$, $\dots$, and pick the best possible answer inside that space at every step, rather than following a fixed formula the way the stationary methods do.

```

function CONJUGATE_GRADIENT(A, b, tolerance, max_iterations):
  x <- zeros(n)
  r <- b - A @ x
  p <- r
  for k in 1 .. max_iterations:
    if norm(r) / norm(b) <= tolerance:
      return x, converged = true
    alpha <- (r . r) / (p . (A @ p))
    x <- x + alpha * p
    r_new <- r - alpha * (A @ p)
    beta <- (r_new . r_new) / (r . r)
    p <- r_new + beta * p
    r <- r_new
  return x, converged = false

```

Conjugate Gradient is valid only on symmetric positive-definite matrices, a restriction that matters a great deal to how its results must be reported, as Section 5.2 explains.

4.7.3 Preconditioned methods

ILU-only, ILU-preconditioned BiCGSTAB, ILU-preconditioned GMRES(30), and a symmetry-dispatched rule that chooses between preconditioned Conjugate Gradient and preconditioned BiCGSTAB depending on whether the matrix is symmetric, round out the sixteen methods. A preconditioned method builds a cheap, approximate factorization M of A , an incomplete LU factorization that deliberately discards the small fill-in entries a complete factorization would keep, and solves $M^{-1}A\vec{x} = M^{-1}\vec{b}$ with a Krylov method instead of solving $A\vec{x} = \vec{b}$ directly, because $M^{-1}A$ behaves far more like the identity than A does on its own.

```

function ILU_PRECONDITIONED_BICGSTAB(A, b, tolerance, max_iterations):
  M <- incomplete_LU_factorization(A)
  apply_M_inverse(v) <- solve M @ w = v for w
  return BICGSTAB(A, b, tolerance, max_iterations,
                  preconditioner = apply_M_inverse)

```

The symmetry-dispatched rule is, on the surface, this project's original proposed contribution, inherited from the group's course proposal. It is presented here purely as a baseline, and the reason it was demoted from a contribution to a baseline is the first episode told in Section 5.

5 Novel Methods: The Design, the Struggle, and the Fix

This section tells the project’s own contribution the way it actually happened, in the order it happened, rather than as a tidy summary written after the fact. Nothing here is reconstructed from memory; every milestone is backed by a script, a data table, or a test that still exists in the repository. The reason for telling it this way is that the mistakes and the false starts are not embarrassing footnotes to the result; they are a large part of the result, and a reader who only sees the polished final pipeline has no way to judge how much of it was luck.

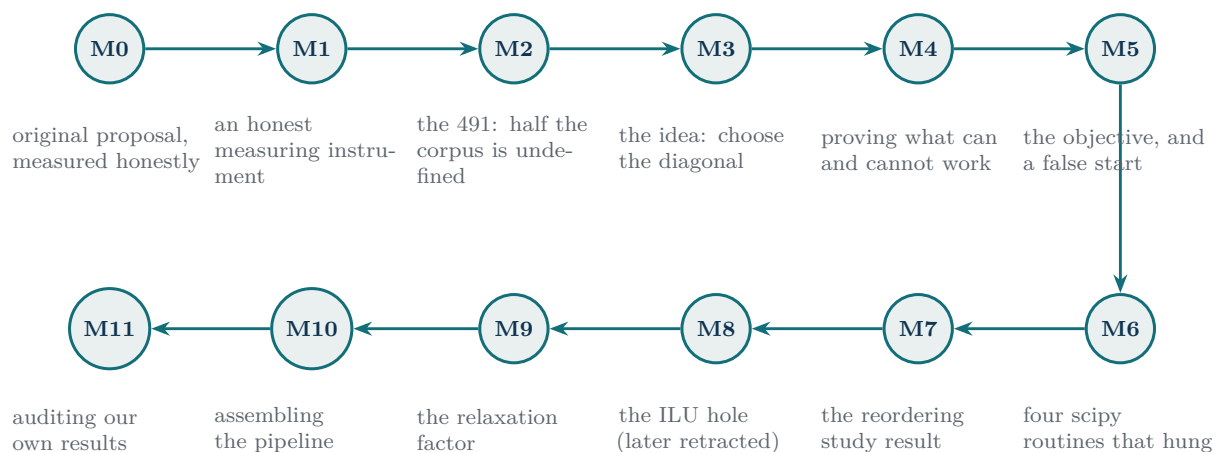


Figure 6: The eleven milestones of the project’s own contribution, in the order they actually happened, read left to right along the top row and then left to right again along the bottom row. Sections 5.1 through 7.7 follow this same order.

5.1 Milestone 0: the original proposal, measured honestly

Before any of the analysis in this report existed, the group’s original course proposal already contained a candidate for a novel contribution: a rule for choosing a solver based on one property of the matrix,

if A is symmetric: use Preconditioned Conjugate Gradient
 else: use Preconditioned BiCGSTAB

with both branches sharing the same ILU preconditioner. We called this rule APK internally and it was written up as the project’s proposed novelty. When we measured it honestly, against the actual literature rather than against our own assumptions, this exact rule turned out to already have a name. It is the default dispatch behaviour described in Barrett et al.’s *Templates for the Solution of Linear Systems* (SIAM, 1994), and it is the built-in default in both PETSc and MATLAB’s backslash operator. It was not something we invented; it was something everybody already does.

Worse, when we measured what the rule contributes on top of simply always using BiCGSTAB, never bothering to check symmetry at all, the answer was close to nothing:

ILU-Krylov (dispatched): 617 of 736 solved
 always ILU-BiCGSTAB: 616 of 736 solved
 net contribution of the dispatch rule: **+1 matrix out of 736**

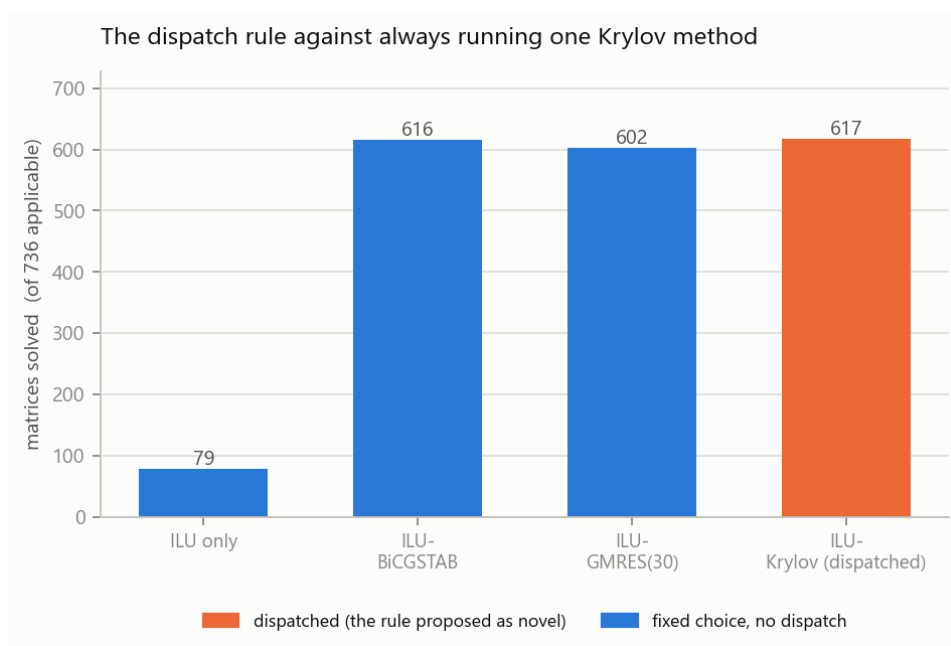


Figure 7: The symmetry-dispatch rule (APK), measured against always using BiCGSTAB regardless of symmetry. The entire measured contribution of the rule is one matrix out of 736.

One matrix, out of 736. That number is the reason this project needed an actual new contribution. The rule is kept in the final benchmark today as `ilu_krylov_dispatched` in `reference_solvers.py`, but honestly labelled as a known baseline rather than a contribution, with a docstring stating plainly what is written above. This was the first lesson of the entire project, and it set the tone for everything that followed: measure everything honestly, including your own idea, before claiming it is new. A full, dedicated treatment of why this rule failed to be novel, and what its measured value actually is, appears again in Section 7.3.

5.2 Milestone 1: building an honest measuring instrument

Before any new idea could be tested fairly, the benchmark itself had to be trustworthy, and the earliest version of the code had several problems, each only found by carefully auditing its own output rather than trusting it.

A solver’s own claim of success was being trusted uncritically: if a routine returned without raising an exception, it was marked successful, even though some of those runs had a relative residual as large as 9.71×10^{12} , wrong by twelve orders of magnitude. Success rates were divided by the size of the whole corpus, even for methods such as Conjugate Gradient that are mathematically defined on only a small, symmetric positive-definite slice of it, which unfairly penalised every specialised method. Iterative refinement, a technique that polishes an already-computed answer with one additional cheap step, was being applied to only one method, which is exactly why that method appeared to be the most accurate; the advantage being measured was the refinement step, not the method underneath it. Gauss-Seidel and SOR were re-analysing their triangular system from scratch on every single iteration, up to ten thousand times per matrix, inflating their measured running time by a factor of eleven to fourteen.

Fixing these was not optional polish; it was the precondition for every later result in this report to mean anything, and Section 7.4 lists all eight mistakes found this way, not only these four, together with exactly what changed once each was fixed. The lasting outcome of this milestone is `metrics.py`, the single place in the codebase that decides whether a solve counted as a success, using eight distinct outcome categories rather than a bare pass or fail, precisely because “the method could not even be attempted” and “the method was attempted and failed” are different facts that a simple binary hides.

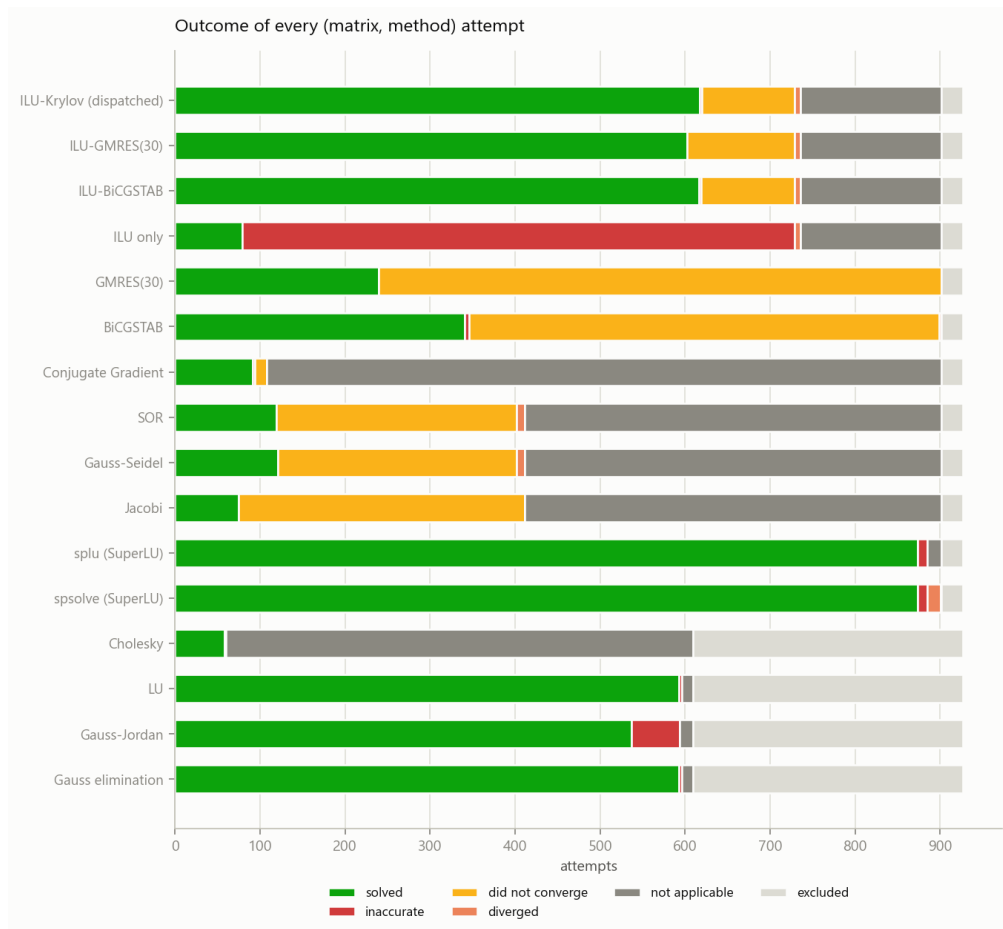


Figure 8: The full breakdown of all 14,832 (matrix, method) attempts across the eight outcome categories the harness distinguishes. Roughly a quarter of all attempts are *not_applicable*, a category that a simple pass/fail measurement would have hidden inside “failed.”

5.3 Milestone 2: the discovery that reshaped the project

Once the harness could be trusted, the first full sweep over the corpus with all sixteen methods produced one number that reoriented the rest of the project. Jacobi, Gauss-Seidel, and SOR all divide by the diagonal entries of A at every step, and if even one diagonal entry is exactly zero, that division is not merely slow or inaccurate; it is undefined, in exactly the sense that $1/0$ is undefined.

491 of 927 matrices (53%) have at least one zero on the diagonal.

This is the single most consequential fact discovered anywhere in this project. Over half the corpus was off-limits to Jacobi, Gauss-Seidel, and SOR before a single iteration was ever attempted, and any comparison that fails to separate “the method could not even try” from “the method tried and failed” is comparing the wrong things. It is what turned the project’s central question from *is Jacobi slow* into *can Jacobi even be made applicable in the first place*, a question with a much larger and more tractable answer space.

5.4 Milestone 3: the idea, choose the diagonal

Once the 491 was understood, an important observation followed almost immediately. The row order of a matrix is simply a list, row one, row two, row three, decided by whoever built the matrix, be it a meshing program, a circuit simulator, or an engineer’s own numbering convention, and that order has nothing to do with which solver will later be applied to it. Crucially, the rows can be reordered without changing the answer at all: if P is a permutation matrix, then $A\vec{x} = \vec{b}$ if and only if $(PA)\vec{x} = P\vec{b}$, so permuting rows is completely free with respect to correctness; it only changes which entries of A end up on the diagonal.

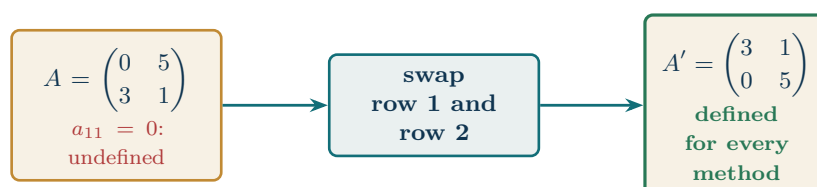


Figure 9: The seed of the entire contribution. Reordering the rows of a matrix changes nothing about the system it represents, but it changes everything about whether the stationary methods can even be attempted.

Same equations, same answer, completely different eligibility for the stationary methods. Choosing which row goes where, before the matrix is ever handed to a solver, is the seed of the entire contribution described in the remainder of this section.

5.5 Milestone 4: proving what can and cannot work

Before searching for a good row order, it was necessary to know which kinds of preprocessing could possibly help, so the search would not waste effort on something mathematically pointless. Three candidates were checked, and two were ruled out by direct algebraic proof rather than trial and error. Scaling every row of A by some factor leaves Jacobi’s iteration matrix $T_J = -D^{-1}(L + U)$ exactly, algebraically unchanged, because the scaling factor cancels in the formula. Scaling every column is a similarity transform, and a similarity transform never changes a matrix’s eigenvalues, hence never changes its spectral radius. Symmetric row-and-column permutation, the classical fill-in-reducing reordering used ahead of direct solvers, was measured directly on real matrices and barely moved the spectral radius at all, from 0.99012 to 0.99011 in the fifth decimal place.

Row permutation on its own is the only one of these operations that changes which numbers sit on the diagonal, and therefore the only one that can change whether Jacobi, Gauss-Seidel, or SOR are even applicable, let alone how quickly they converge. This proof is what turned “try shuffling the rows” from a guess into a principled, justified search: the search was restricted to row permutations specifically because it could be shown that nothing else was worth searching over.

5.6 Milestone 5: building the objective, and a false start

Given that row permutation is the only lever, the next question is which permutation. There are $n!$ possible orderings for an n -row matrix, far too many to try one at a time, so the choice has to be posed as an optimization problem: pick one entry per row, and one per column, to occupy the diagonal, an *assignment problem*, which can be solved exactly rather than approximately.

The existing, industry-standard tool for this is MC64 (Duff and Koster, 1999), which picks the diagonal maximizing the product of the absolute diagonal values. This is an excellent objective for direct solvers, since a large diagonal magnitude means the elimination process is numerically stable, but MC64 was never designed with iterative convergence in mind: a diagonal entry of size 1,000 looks excellent in isolation, but if the remainder of that row sums to 50,000, the row is still poor for Jacobi or Gauss-Seidel, and MC64 has no way to see that, since it only ever inspects the single diagonal entry, never the rest of the row.

The objective proposed in this project looks at the whole row instead. For row i with diagonal candidate a_{ii} , define the row ratio

$$\text{ratio}_i = \frac{\sum_{j \neq i} |a_{ij}|}{|a_{ii}|}$$

A ratio below 1 for every row is exactly the classical condition of strict diagonal dominance, a sufficient condition that guarantees both Jacobi and Gauss-Seidel converge. The ratio is therefore not an arbitrary score; it is the precise quantity the classical convergence theorem is stated in terms of, and this project optimizes it directly, which, as far as we could determine, nobody had previously done for this specific purpose. Minimising the worst such ratio across all rows is called the **bottleneck objective**, because it is a bottleneck assignment problem, a close relative of MC64's maximum-weight assignment problem with a different scoring rule.

A second candidate, called **min-sum**, minimises the total of all the ratios rather than only the worst one, and was implemented as a natural point of comparison. Working out the algebra by hand revealed something the group did not expect: since $1 + \text{ratio}_i = (\text{row } i\text{'s total})/|a_{ii}|$,

$$\sum_i \log(1 + \text{ratio}_i) = \sum_i \log(\text{row totals}) - \sum_i \log |a_{ii}|$$

and the first term on the right does not depend on the permutation at all. Minimising the left-hand side is therefore exactly the same optimization as maximising $\sum_i \log |a_{ii}|$, which is precisely MC64's own objective. This was verified directly on 193 randomly generated matrices: min-sum and MC64 produced an identical objective value on every single one. Min-sum was not a genuine second alternative; it was MC64 under another name. This finding is now written permanently into the `minsum_permutation` function's docstring in `reordering.py`, and a dedicated automated test, `test_minsum_is_mc64_under_another_name`, exists specifically so nobody rediscovers it the hard way. The real competition, in the end, is between two genuinely different objectives: MC64, the established one, and bottleneck, the one this project proposes.

5.7 Milestone 6: four scipy routines that could not be trusted

This milestone cost more real engineering time than any other part of the project, and is worth recounting in full for what it teaches about trusting a library routine before measuring it. The first implementation of the bottleneck search used two ready-made scipy functions built specifically for matching problems of this kind, `min_weight_full_bipartite_matching` and `maximum_bipartite_matching`. Four separate times, in four separate ways, they hung on real matrices from this corpus.

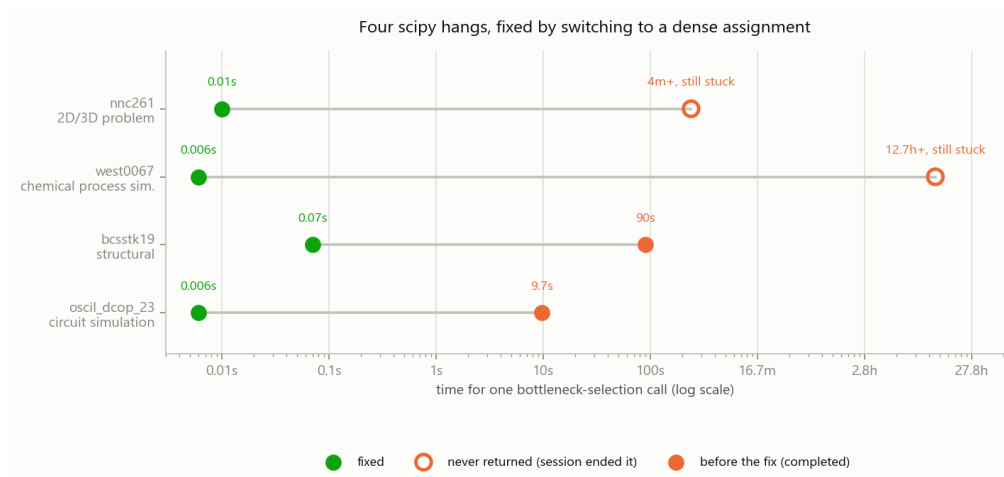


Figure 10: The four scipy routine hangs encountered during development, the matrix that exposed each one, and the wall-clock cost of the fix that replaced the hanging routine with a dense assignment solve.

On `nnc261`, using raw ratio values as matching weights, the routine had not returned after 240 seconds. After switching to a numerically safer weight formulation to fix that first problem, the very same routine hung on a completely different and much smaller matrix, `west0067`, only 67 rows. The other routine, `maximum_bipartite_matching`, took 7.2 seconds for a single call on `bcsstk19`, 817 rows, and the bottleneck search needs roughly ten such calls per matrix; it was, unhelpfully, slowest exactly when a matching did exist, the common case. After both original routines were replaced, a memory-saving size cap meant the largest matrices were still quietly routed through the same broken code path, and a fourth hang appeared on `rw5151`, 5,151 rows, caused by the very safety limit meant to prevent this problem.

The fix, each time, was the same idea: convert the matching problem into a dense assignment problem and hand it to `scipy.optimize.linear_sum_assignment`, an entirely different and, at these matrix sizes, extremely fast and reliable algorithm. Where the original routines took over ninety seconds or never returned at all, the dense replacement finished in a fraction of a second, in one measured case more than 1,600 times faster. The deeper lesson is that a hang cannot be interrupted from inside a running program; the only real defence is to test every matrix in the corpus locally before spending any cloud compute time on it. This is exactly what `preflight_reordering.py` does: it runs every objective over the entire corpus on an ordinary computer first and refuses to let anything proceed to the cloud until all 930 matrices pass without a single hang. The eventual full preflight run completed all 930 matrices in 66.5 minutes with zero problems, proof that the fix actually worked rather than an assumption that it did.

5.8 Milestone 7: the reordering study result

With the search made reliable, the actual experiment could finally be run: for every matrix, for each of the three stationary methods, try solving under three conditions, the row order as given, the row order chosen by MC64, and the row order chosen by this project’s full selector, which computes every candidate diagonal and keeps whichever gives the smallest estimated $\rho(T_{GS})$.

systems solved, as given:	317
systems solved, MC64:	514
systems solved, our selector:	516

199 systems rescued in total. Zero lost, since “change nothing” is always one of the selector’s own candidates.

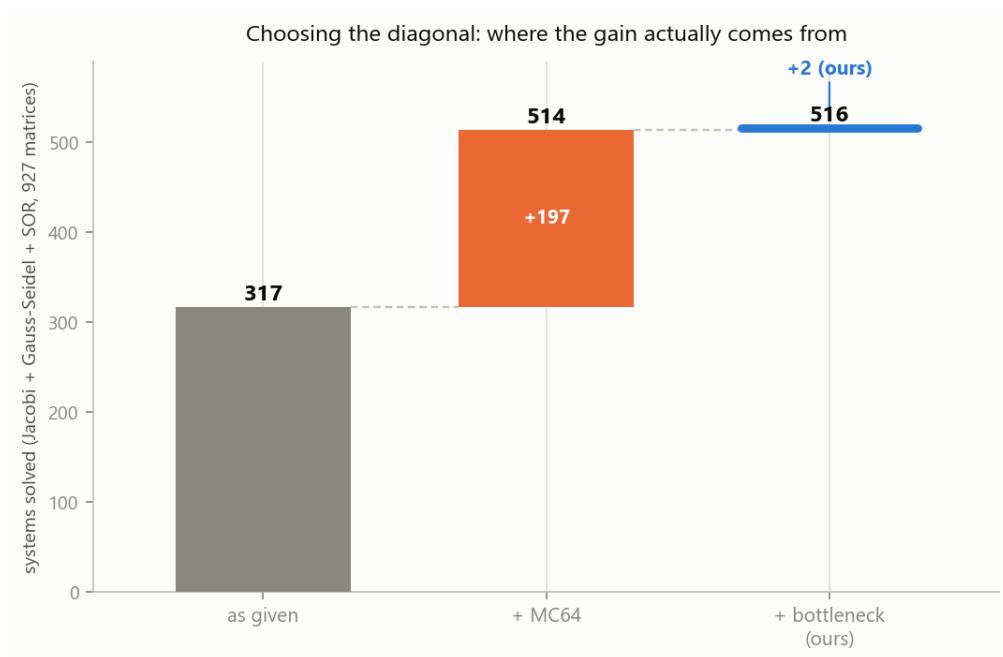


Figure 11: Systems recovered, incrementally, as each preprocessing step is added on top of the previous one, per stationary method.

Almost all of that gain, however, belongs to MC64, not to the bottleneck idea. Of the 199 rescued systems, 195 are rescued by both objectives together, 4 by bottleneck alone, and 2 by MC64 alone, a net gain for the new objective over the established one of only two systems. This mirrors Milestone 0 closely: the idea of reordering is genuinely powerful, but the specific choice of objective barely changes the outcome, at least by this particular measure. On the quantity bottleneck was actually built to minimise, the worst row ratio itself rather than the count of systems solved, it wins outright: across the 873 matrices where both objectives produce a finite ratio, bottleneck is strictly better on 367, MC64 is strictly better on precisely 0, and 506 are tied.

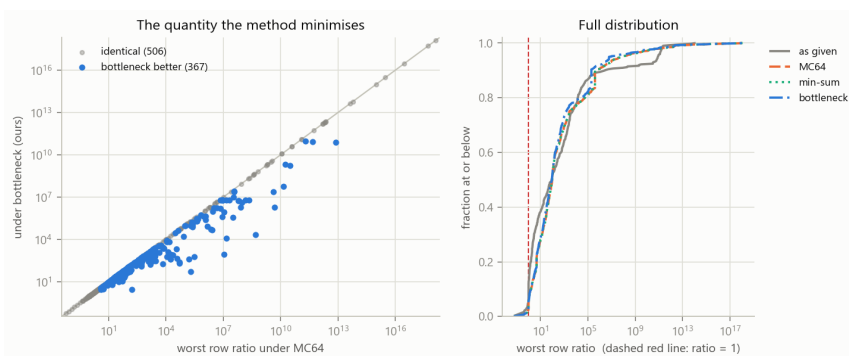


Figure 12: Every matrix's worst row ratio under MC64 against the same quantity under bottleneck. Every point sits on or below the diagonal line; bottleneck never loses on the quantity it was designed to minimise.

And the theoretical guarantee the objective was built around, driving the worst row ratio below 1, turns out to be unreachable on this corpus: 43 matrices were already below the threshold before any reordering at all, and not one further matrix can be pushed under it by any row permutation whatsoever. The theorem is true, and, on this data, empty. Why a result that wins 367 to 0 on

its own target quantity converts only two additional systems is a question this project came back to directly in its final audit, and the full explanation is given in Section 7.5.

5.9 Milestone 8: the ILU hole (later retracted, see Section 7.7)

While writing up the reordering result, a related question arose: does the same zero-on-the-diagonal problem also disable the preconditioned methods, not only the three stationary ones? The routine that builds an ILU preconditioner tries a proper incomplete factorization at several settings and, if all fail, falls back to a cruder approximation, simple diagonal scaling, which also needs a nonzero diagonal, so a zero on the diagonal closes every door at once, taking the entire preconditioned family down together.

At the time, we measured that on 166 matrices where no ILU preconditioner could be built at all, reordering made 150 of the 166 constructible, and reported this as a second result of the reordering pipeline. **This claim did not survive a later audit and has been retracted**, for reasons given in full in Section 7.7; it is recorded here, in its original place in the timeline, because the chronology of the mistake, and how it was caught, is itself part of what this report was asked to explain, and repeating it only in the failure-analysis section would misrepresent when it actually happened.

5.10 Milestone 9: the relaxation factor, chosen per matrix

SOR has one tuning parameter Jacobi and Gauss-Seidel do not: the relaxation factor ω . Throughout the main sweep, ω was fixed at 1.25 for every matrix in the corpus, one number applied uniformly regardless of what the matrix actually looked like. Measuring whether that fixed choice was even a good idea revealed it was wrong in both directions at once: plain Gauss-Seidel, equivalent to $\omega = 1$, solves 11 systems that SOR at the fixed $\omega = 1.25$ does not, while SOR at $\omega = 1.25$ solves 9 systems that plain Gauss-Seidel does not. Twenty systems in the corpus, in other words, have their outcome decided entirely by the choice of ω , which is the ceiling of what any smarter choice could possibly recover.

Young's 1950 formula gives the mathematically optimal relaxation factor from the spectral radius of the Jacobi iteration matrix:

$$\omega^* = \frac{2}{1 + \sqrt{1 - \rho(T_J)^2}}$$

```
function ESTIMATE_RHO_JACOBI(A, power_iterations = 60):
  v <- a random unit vector
  for step in 1 .. power_iterations:
    v <- v - (A @ v) / diagonal(A)
    v <- v / norm(v)
  return norm(v)

function OPTIMAL_OMEGA(rho):
  if rho is not usable, or rho >= 1:
    return 1.0
  return 2 / (1 + sqrt(1 - rho * rho))
```

```
function SOR_ADAPTIVE(A, b, tolerance, max_iterations):
  rho    <- ESTIMATE_RHO_JACOBI(A)
  omega  <- OPTIMAL_OMEGA(rho)
  return SOR(A, b, omega, tolerance, max_iterations)
```

$\rho(T_J)$ is estimated cheaply with roughly sixty power-iteration steps and plugged into Young's formula per matrix, individually, instead of one fixed number for every matrix. Young's formula is exact only for consistently ordered matrices with property A; most of this corpus does not belong to that class, so applying the formula anyway is a measured choice, not an assumed one.

Gauss-Seidel ($\omega = 1$):	122 solved
SOR, fixed $\omega = 1.25$:	119 solved
SOR, ω chosen per matrix:	127 solved

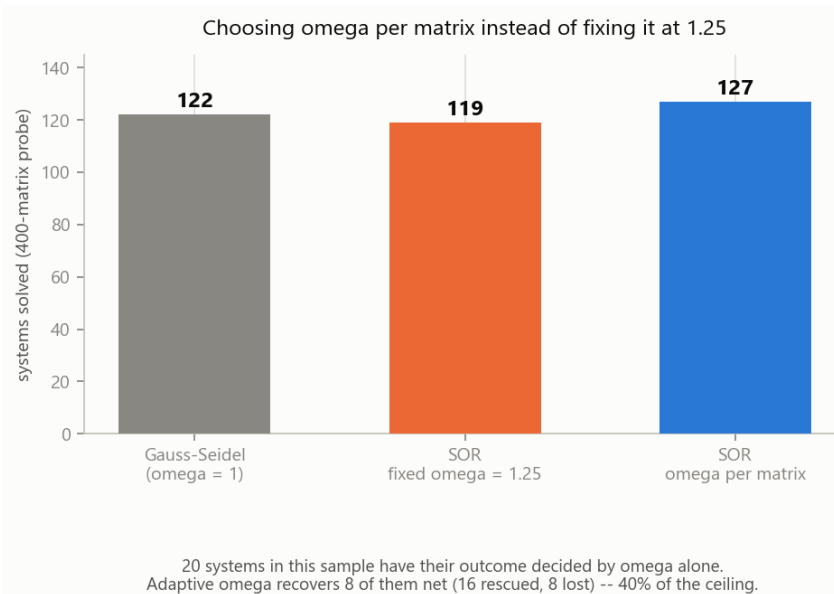


Figure 13: Systems solved by Gauss-Seidel, fixed- ω SOR, and adaptive- ω SOR, on the 400-matrix probe where a usable diagonal exists.

Eight net systems recovered out of the twenty-system ceiling identified above, forty percent of everything that was even possible to gain this way.

5.11 Milestone 10: assembling the pipeline

Putting Milestones 7, 8, and 9 together gives the final structure of the contribution: a two-step preprocessing pipeline that runs in front of an otherwise completely unmodified solver.

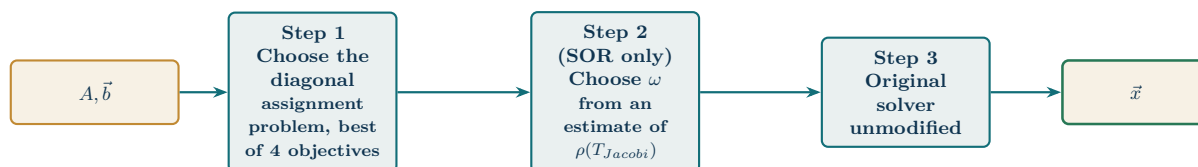


Figure 14: The complete preprocessing pipeline, reused here directly from the project’s presentation deck. Steps 1 and 2 never touch the solver itself: $A'\vec{x} = \vec{b}'$ solves the original system exactly, under a relabelling of the rows, so nothing needs to be undone once the solver returns.

Every arm of this pipeline is measured separately as well as together, because “the pipeline helps” is not, by itself, a real result; which specific step helps, and by how much, is the real result. The full-corpus run measuring every pipeline arm together completed on all 927 matrices in 413.6 minutes with no hang and no budget stop, and it confirmed two things that the smaller probes in Milestones 8 and 9 could not show on their own.

Reordering alone already takes SOR from 119 to 184 (Milestone 7); adding adaptive omega on top takes it to 208, a 75% increase over the as-given baseline, stronger than reordering alone’s 55%. This second step is not loss-free the way the first one is: relative to step 1 alone it gains 36 and loses 12, because there is no “leave ω alone” fallback the way “leave the diagonal alone” is always one of step 1’s own candidates.

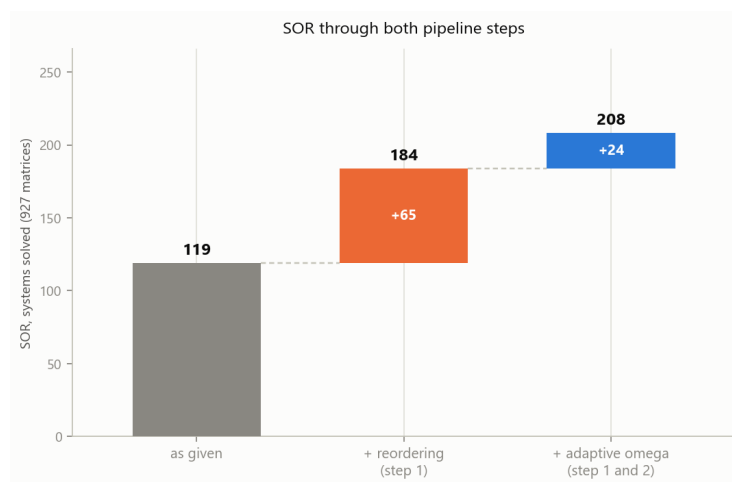


Figure 15: SOR solved counts through both pipeline steps, measured on the full 927-matrix corpus in one run.

The second confirmation, reordering measured in front of ILU across the entire corpus rather than only the 166-matrix subset from Milestone 8, is reported in this report only with the caveat attached in Section 7.7: it rests on the same defect in the ILU construction routine that Milestone 8’s headline claim did, and a corrected re-measurement was queued but has not completed as of the date of this report. Both the original number and why it cannot yet be cited as final are discussed there in full.

6 Results

This section reports the measured numbers behind every claim made so far, restricted to results that do not depend on the flawed ILU construction routine discussed in Section 7.7. Results that do depend on it are deferred entirely to that section, where they belong alongside the audit that found the defect.

6.1 The headline comparison across sixteen methods

Table 3 reports applicability and conditional success as two separate numbers for every method, never multiplied together, for the reason given in Section 5.2: Conjugate Gradient, for instance, is applicable to only 11.7% of the corpus, its symmetric positive-definite slice, but solves 84.3% of what it is applicable to, a fact a single blended percentage would hide entirely.

Table 3: Applicability and conditional success for all sixteen methods, corpus of 927 matrices.

Method	Applicable	Solved	Applicability	Cond. success
Gauss elimination	596	593	64.3%	99.5%
LU	596	593	64.3%	99.5%
splu (SuperLU)	885	874	95.5%	98.8%
spsolve (SuperLU)	901	874	97.2%	97.0%
Cholesky	60	58	6.5%	96.7%
Gauss-Jordan	594	537	64.1%	90.4%
Conjugate Gradient	108	91	11.7%	84.3%
ILU-Krylov (dispatched)	736	617	79.4%	83.8%
ILU-BiCGSTAB	736	616	79.4%	83.7%
ILU-GMRES(30)	736	602	79.4%	81.8%
BiCGSTAB	902	341	97.3%	37.8%
Gauss-Seidel	411	121	44.3%	29.4%
SOR ($\omega = 1.25$)	411	119	44.3%	29.0%
GMRES(30)	902	240	97.3%	26.6%
Jacobi	411	75	44.3%	18.2%
ILU only	736	79	79.4%	10.7%

The ILU-family rows of this table are subject to the caveat in Section 7.7 and are almost certainly a slight underestimate of what the preconditioner can actually do; the direction of the underlying defect makes construction harder than it needs to be, not easier, so these numbers are a conservative floor rather than a misleading ceiling. A sparse direct solver wins outright on raw reliability: `spsolve` solves 874 of 927 matrices, against 617 for the best iterative configuration, a comparison the project’s very first, uncorrected sweep never made available, since applicability and success were still being conflated at that stage.

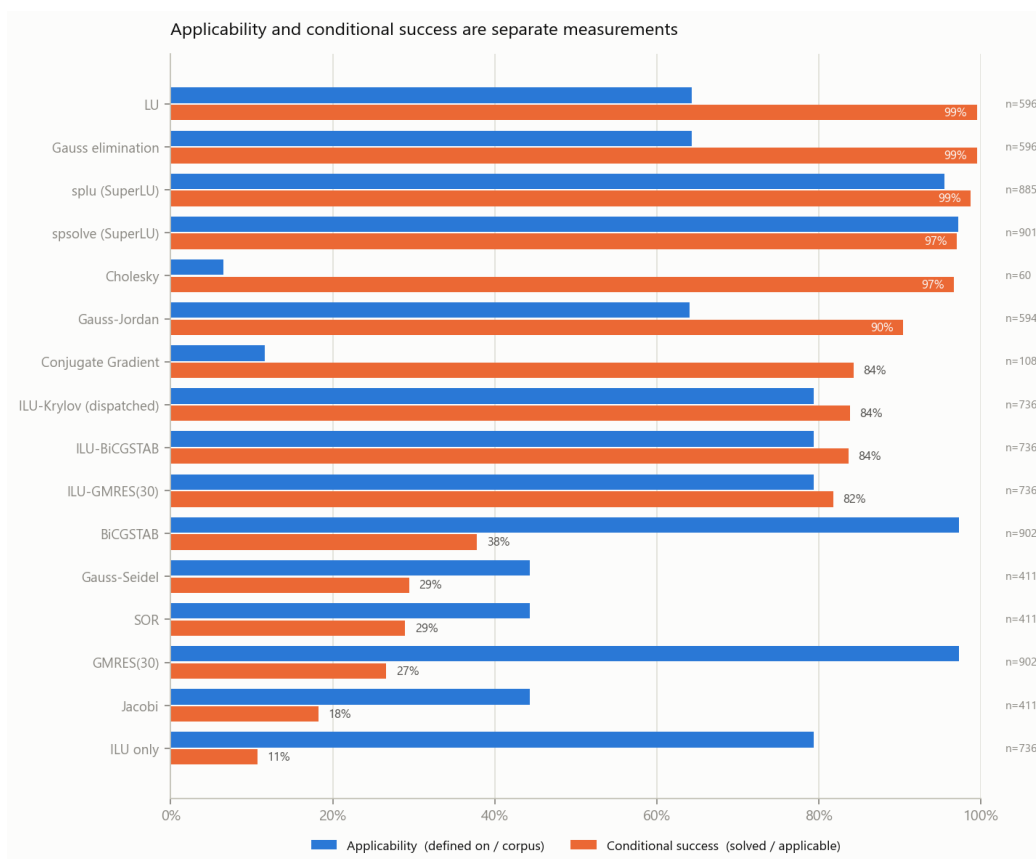


Figure 16: Applicability against conditional success for all sixteen methods. Methods in the upper right are both broadly applicable and reliable once applicable; ILU-only sits in the lower right, broadly applicable but rarely successful on its own.

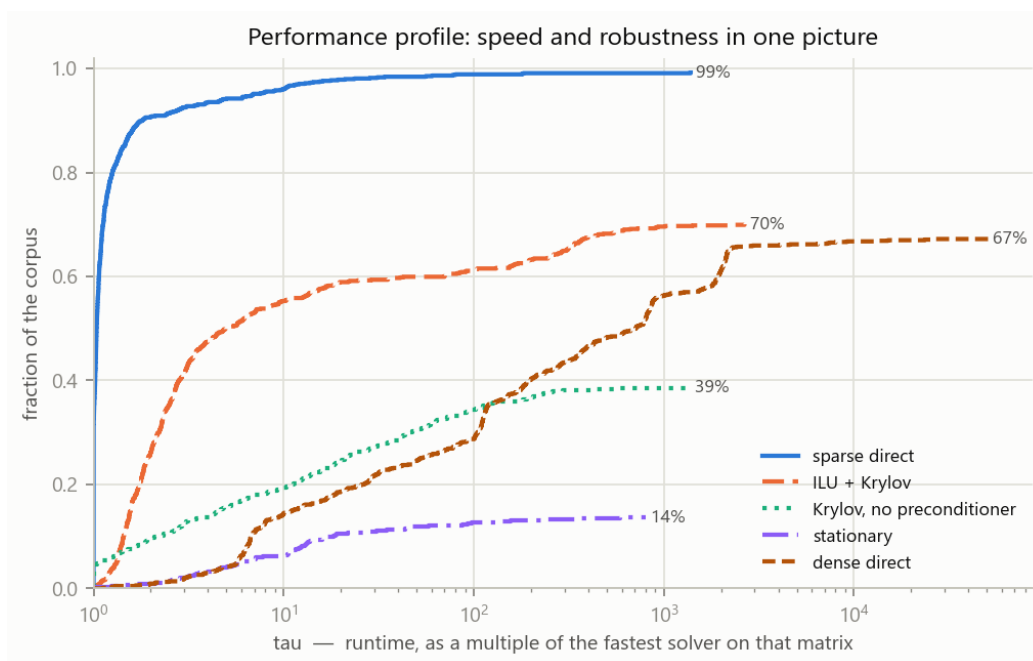


Figure 17: A performance profile: for each method, how often it is the outright fastest solver on a matrix it can solve, against how much of the corpus it can solve at all.

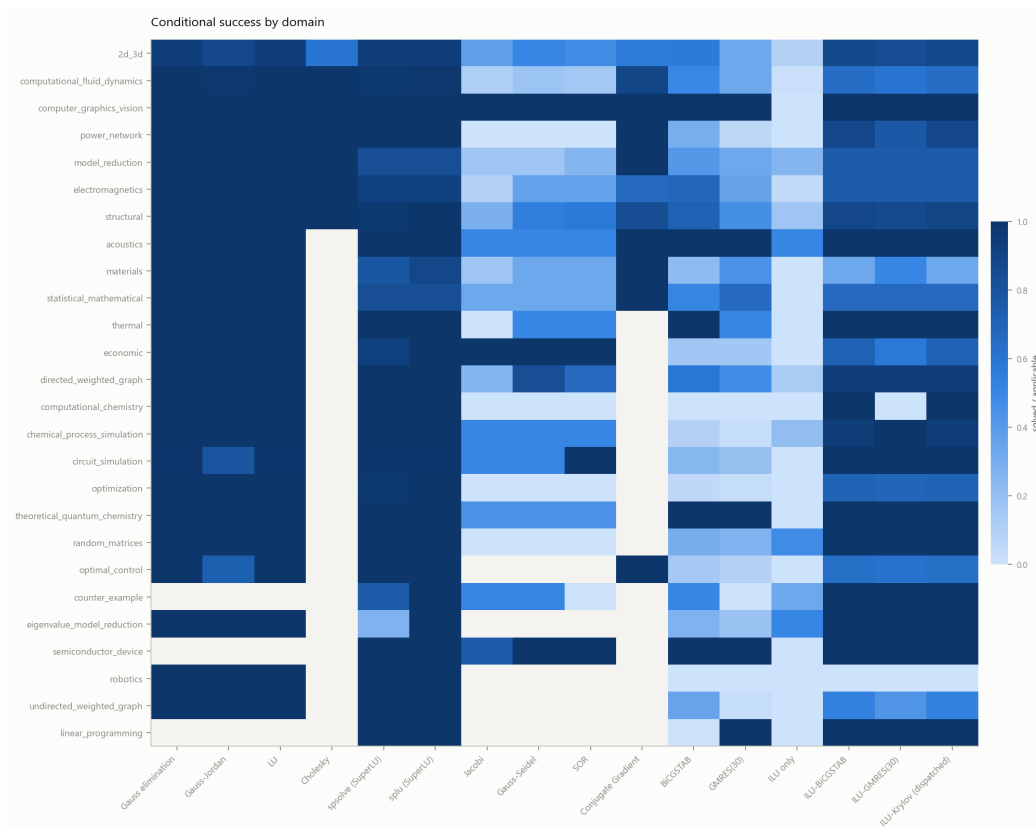


Figure 18: Conditional success rate by method and by domain. Some domains are far friendlier to the stationary methods than others; circuit-simulation matrices, in particular, are disproportionately represented among the zero-diagonal failures of Section 5.3.

6.2 Cost and scaling

Cost is measured in matrix-vector products rather than wall-clock seconds, since shared cloud hardware timing is not reproducible and iteration counts are not comparable across methods, one ILU-preconditioned step does the work of many Jacobi steps.

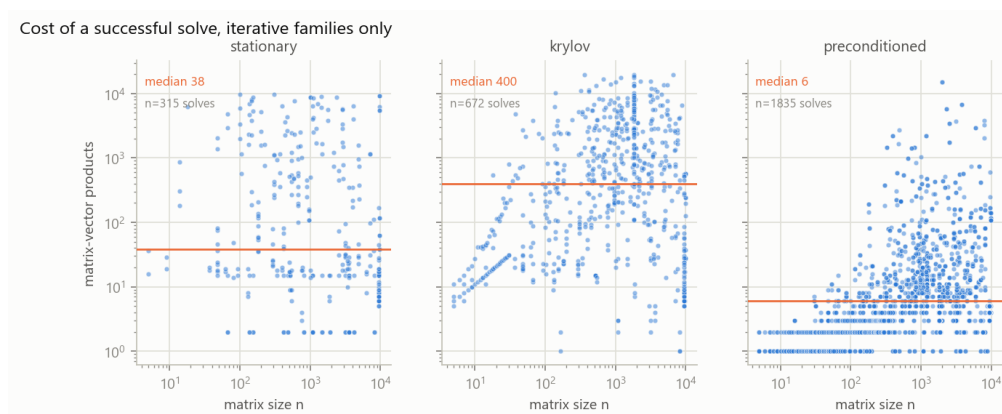


Figure 19: Cost, in matrix-vector products, for each method's successful solves. Direct methods pay their entire cost up front; iterative methods trade a larger number of cheap steps for that same up-front cost.

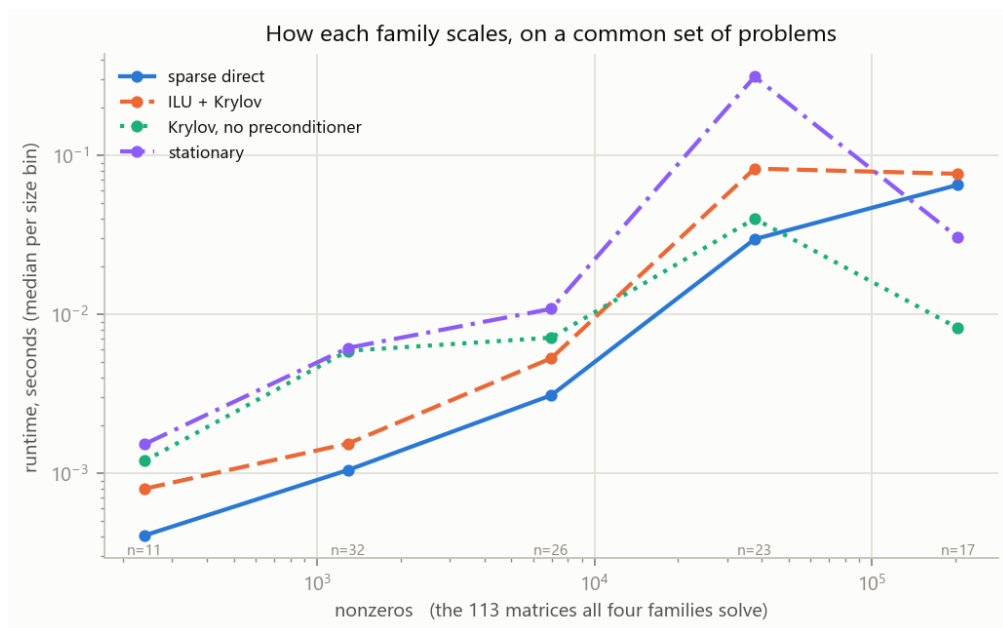


Figure 20: How measured running time grows with matrix size, by method family.

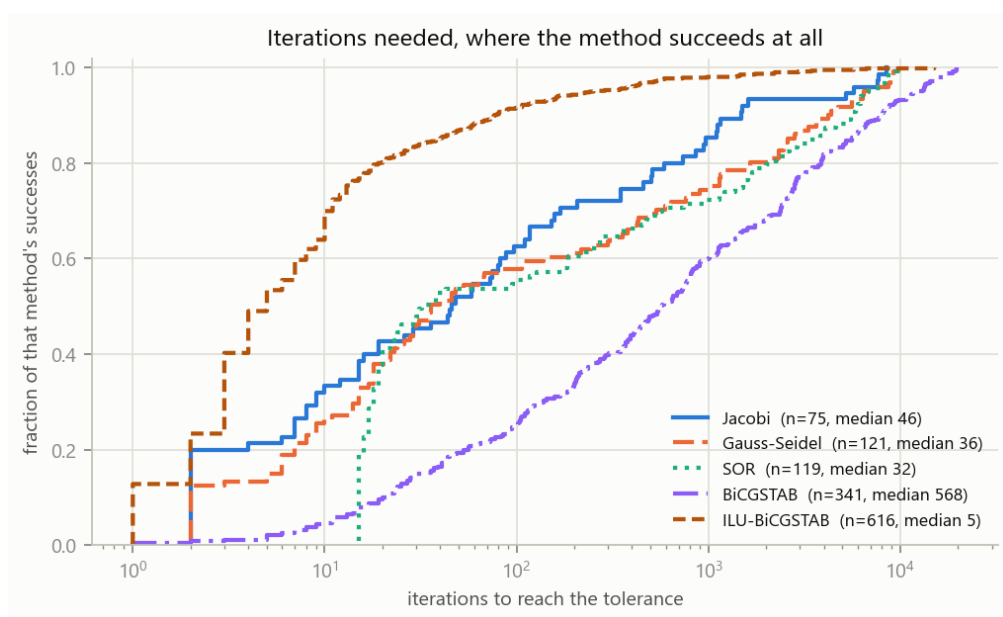


Figure 21: Iteration counts needed for convergence, among successful solves, by method.

6.3 The base paper's own test, on real data

Restricting to the 411 systems where both Jacobi and Gauss-Seidel are even defined, the comparison the base paper itself makes looks like this on real data:

Table 4: Convergence outcomes for Jacobi and Gauss-Seidel, restricted to the 411 systems where both are applicable.

Outcome	Count
Both converge	75
Only Gauss-Seidel converges	46
Only Jacobi converges	0
Neither converges	293

Khrapov and Volkov report 1,095 only-Jacobi cases among their own random matrices at $n \leq 5$; on this corpus, at realistic sizes, that case never once occurs. On the 100 self-generated random matrices built specifically to reproduce the base paper’s own validation approach at larger scale, the picture is starker still: neither Jacobi nor Gauss-Seidel solves a single one of the 100, while a direct solver solves all 100, and the spectral radius of the random-matrix iteration matrices grows roughly in proportion to matrix size, so the paper’s own validation method stops working almost as soon as the matrices are made larger than the paper itself tested.

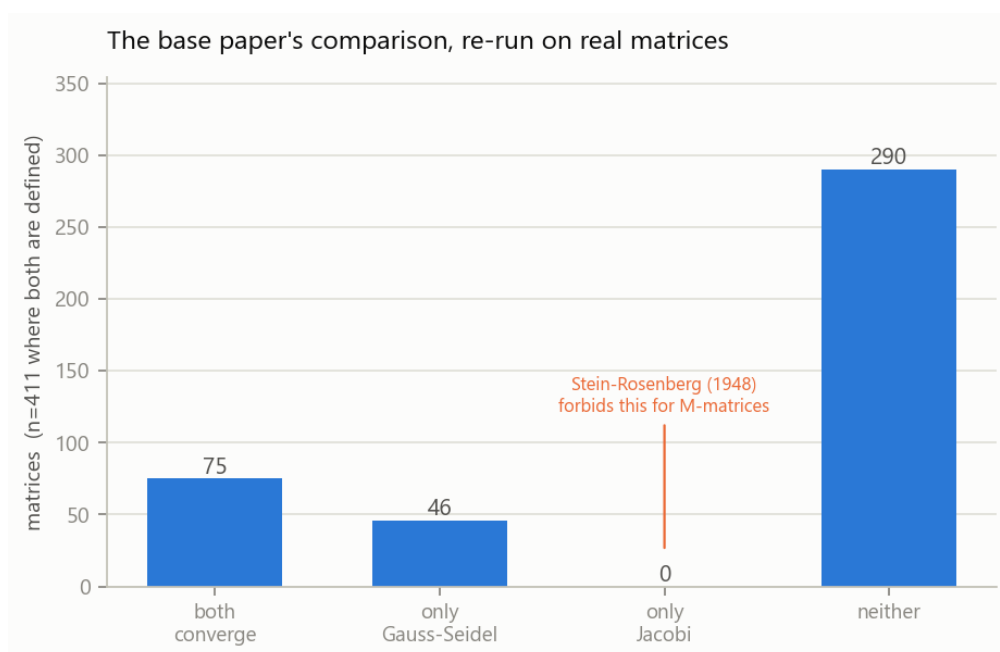


Figure 22: The base paper’s own question, Jacobi against Gauss-Seidel, answered on real matrices rather than random ones.

6.4 Spectral prediction, validated

The textbook criterion, $\rho(T) < 1$ implies convergence, agrees with the actually observed outcome on 99.3% of Jacobi attempts and 98.2% of Gauss-Seidel attempts across the 905 matrices where the spectral radius could be computed exactly. The one Jacobi disagreement worth naming individually is `cdae6`, already introduced in Section 3.4: $\rho(T_J) = 0.717$, provably convergent, yet the residual peaks at 44,764 times its starting value at iteration 73 before converging at iteration 178, a transient large enough that an overly strict divergence guard could mistake it for genuine divergence. Once this single case is accounted for, the theory’s `converges` prediction for Jacobi is correct on all 71 of 71 cases it applies to.

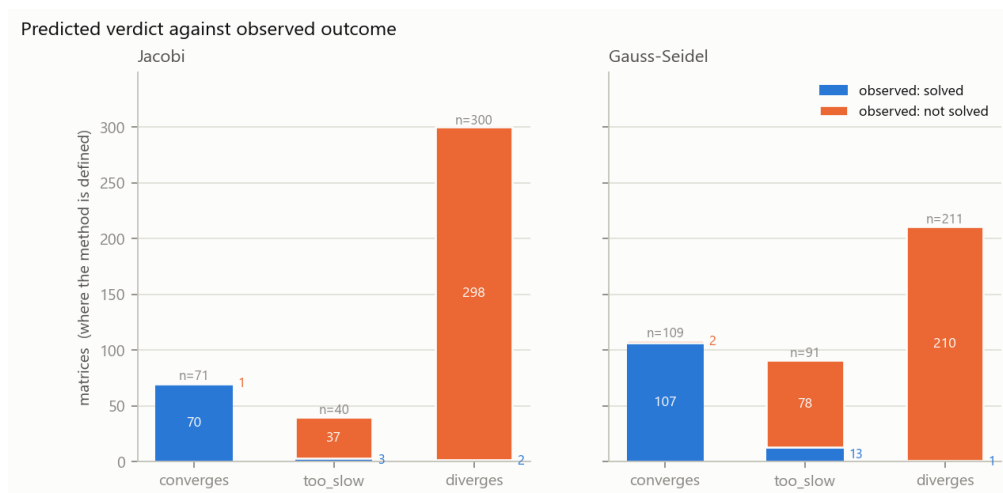


Figure 23: Predicted convergence from the spectral radius against the observed outcome, for Jacobi and Gauss-Seidel.

6.5 Refinement, given to every method

Applying one pass of iterative refinement identically to every method, rather than to a single favoured one as in the earliest version of this project, changes the accuracy ranking outright: Jacobi’s median relative forward error falls from 1.893×10^{-8} to 6.985×10^{-16} , making it, after refinement, the single most accurate method in the entire benchmark, a factor of one hundred better than the dispatched ILU method’s 7.111×10^{-14} . This comes at a real cost in matrix-vector products, 59 without refinement against 279 with one pass, a cost that would have been invisible under the earliest version of the harness, which measured accuracy for one method at a time rather than as an orthogonal, universally applied factor.

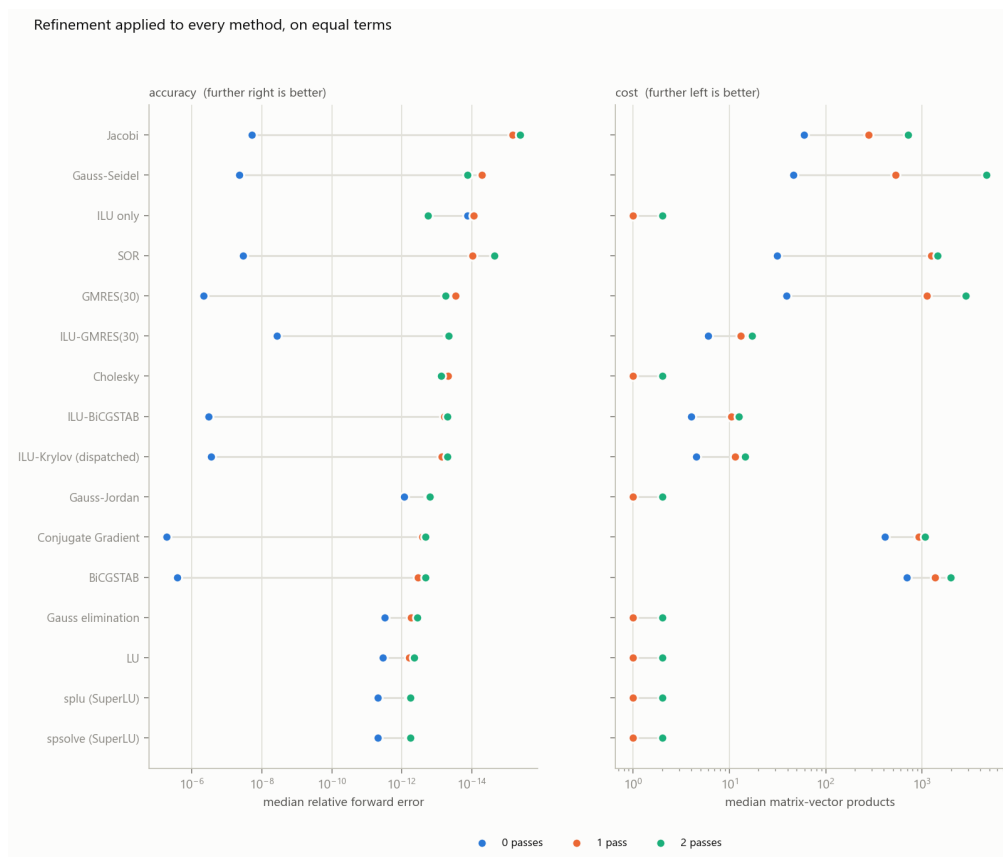


Figure 24: Median relative forward error at zero, one, and two refinement passes, by method.

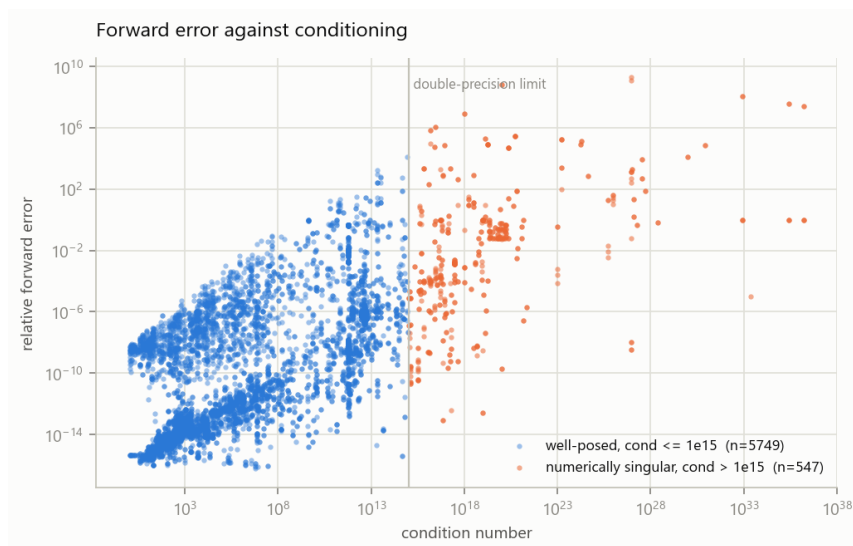


Figure 25: Accuracy, split by whether the underlying matrix is numerically ill-conditioned, across methods.

6.6 The reordering pipeline, on the stationary methods

The headline contribution of this project, restricted to the parts that do not touch ILU construction, is a preprocessing pipeline that raises the success rate of the three stationary solvers substantially, with every gain measured on the same run as its own baseline.

Table 5: Systems solved by each stationary method, as given and with the full preprocessing pipeline, 927-matrix corpus.

Method	As given	With pipeline	Change
Jacobi	76	124	+63%
Gauss-Seidel	122	208	+70%
SOR	119	184	+55%

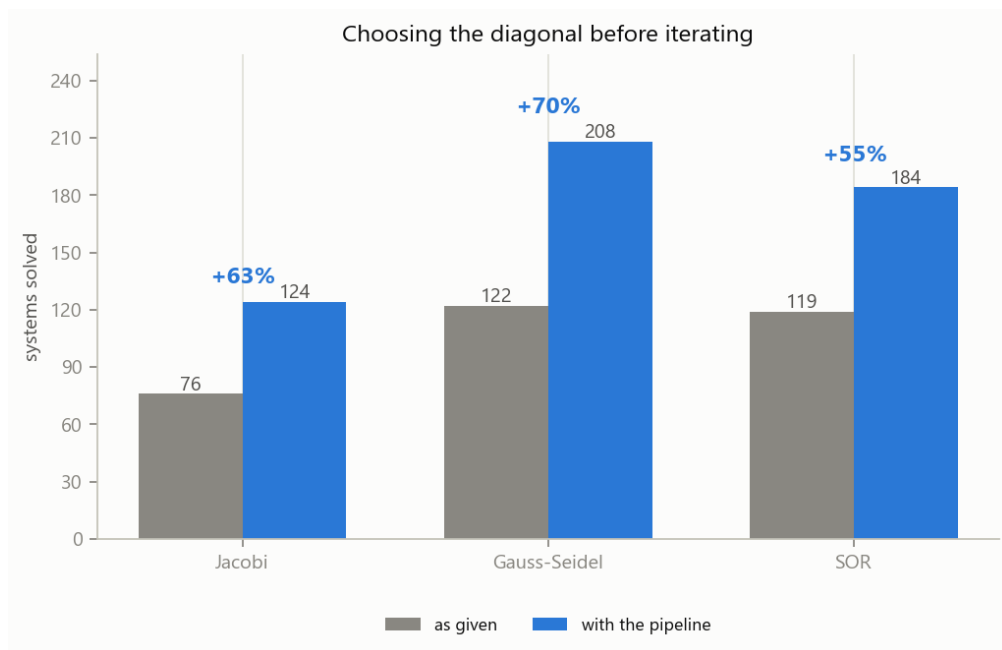


Figure 26: The headline pipeline result: systems solved as given against systems solved with the full pipeline, per stationary method, with the percentage gain printed above each pair.

One hundred ninety-nine systems recovered in total, none lost, since the selector described in Section 5.6 always includes “change nothing” among its own candidates and so can never score worse than doing nothing at all. Precisely how much of that gain comes from the diagonal-selection step against the relaxation-factor step, and precisely what that gain is actually made of, is the subject of the failure analysis in Section 7, since understanding the composition of a positive result and diagnosing the failure of a related one turned out, in this project, to be the same investigation.

7 Failure Analysis

Our course teacher specifically asked for a dedicated treatment of why a given solver fails, and why, together with what novel remedy was proposed in response, for both the classical methods from the base paper and the methods this project designed itself, even where some of this ground has already been touched on earlier in the report. This section is that treatment, gathered in one place. It is organised as a sequence of specific failures, each stated as plainly as we could manage: what failed, why it failed, what we tried in response, and, where a remedy exists, how much of the failure it actually closes. Two of the entries below are failures of the project's own methodology, not of a solver, because an honest failure analysis of this project has to include the mistakes the project itself made, not only the mistakes made by the matrices it was testing.

7.1 Why Jacobi, Gauss-Seidel, and SOR fail: undefined before they start

The most basic failure mode of the three classical stationary methods on this corpus is not slowness or inaccuracy; it is that the method cannot be attempted at all. Section 5.3 already established the mechanism: each method divides by the diagonal entries of A , and 491 of 927 matrices, 53% of the corpus, carry at least one zero on that diagonal. This is a structural failure, decided entirely by which row of the matrix happened to be listed as which row, an accident of the software that produced the matrix rather than any property of the underlying physical system. The remedy this project designed in response is the diagonal-selection pipeline of Section 5.6, and it converts the bulk of this exact failure mode into applicability: of the 199 systems the pipeline recovers, 182, or 91%, are cases where the diagonal was empty beforehand and a permutation supplies a nonzero one, a finding developed fully in Section 7.8 below. This is the one failure mode in this entire section that the project's own remedy addresses almost completely, because supplying a nonzero diagonal is exactly what a perfect matching does whenever one exists at all, and a perfect matching exists on this corpus far more often than not.

7.2 Why they still fail when defined: the theorem wall

The remaining failure mode is harder, and it applies even to matrices where the diagonal was never a problem. Table 6 recomputes the classical convergence-guarantee classes against actual observed success, restricted correctly to matrices whose underlying system is itself solvable (excluding structurally singular matrices and load failures, a correction to an earlier pass of this same analysis that had wrongly left those in the denominator).

Table 6: Success rate by classical hypothesis class, restricted to solvable matrices.

Class	Matrices	Jacobi	Gauss-Seidel	SOR
Strict diagonal dominance	43	88%	88%	88%
Weak diagonal dominance	46	83%	83%	85%
H-matrix	83	77%	81%	76%
M-matrix	34	65%	76%	76%
L-matrix	38	58%	71%	68%
Symmetric positive definite	102	27%	51%	55%
Property A (proxy)	117	24%	29%	28%
Symmetric	298	13%	22%	24%
No theorem applies	679	0%	3%	3%

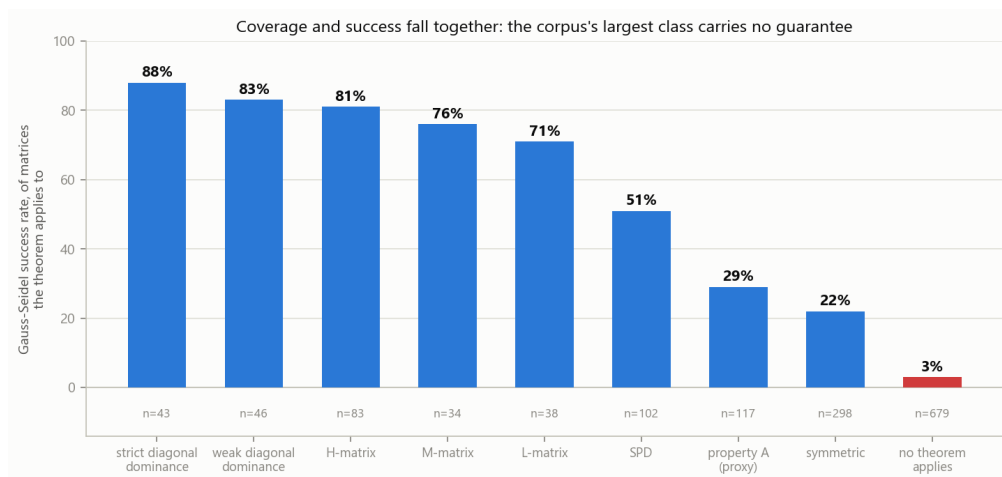


Figure 27: Gauss-Seidel success rate by classical hypothesis class. The bar for “no theorem applies,” at 3%, is the largest single population in the corpus and the smallest success rate.

Where some classical theorem applies at all, across 223 solvable matrices, Gauss-Seidel solves 46%; where none applies, across 679 matrices, it solves 3%. The classical theorems are never wrong: every strictly diagonally dominant matrix in the corpus does have $\rho(T_{GS}) < 1$, with a maximum observed value of 0.999999. But five of those guaranteed-convergent matrices still fail inside the iteration budget, every one of them sitting at $\rho > 0.99$, and this gap between converges and converges usefully, first introduced in Section 4.5, is what defeats every permutation-based remedy attempted in this project, as the next three subsections show in turn.

7.3 Why the original novel proposal, APK, was not novel

Section 5.1 already told the story; this subsection states its failure mode plainly, as the failure analysis this section was asked to provide. The rule failed to be a contribution for two independent reasons, and it is worth separating them because they call for different lessons. First, it failed a literature check: symmetry-dispatched preconditioned Krylov selection is the documented default behaviour in Barrett et al.’s *Templates* (1994) and the built-in default in both PETSc and MATLAB, so the rule was not new regardless of how it performed. Second, and independently, it failed a measurement check: even judged purely on its own merits, ignoring the literature entirely, it contributes one solved matrix out of 736 over the naive baseline of always choosing BiCGSTAB. Either failure alone would have been enough to disqualify it as a contribution; having both at once removed any ambiguity. The remedy was not a fix to the rule; it was demoting the rule to an honestly labelled baseline and building an actual new objective, described starting in Section 5.4, in its place. The lesson this failure taught, that every later milestone in this project tried to apply, is to check both the literature and the measured effect size before calling anything novel.

7.4 Why the first benchmark harness produced false results

A benchmark that lies to itself before any solver is even compared is a different kind of failure from a solver that fails on a hard matrix, but it is no less important to a failure analysis, since every later result in this report depends on this one having been caught and fixed. Table 7 lists all eight defects found in the earliest version of the project’s own measuring instrument, each one changing a headline number once corrected.

Table 7: Eight methodological defects found in the project’s own harness, and their measured effect once fixed.

Defect	Effect of the fix
Trusted a solver’s own success flag	57 false successes caught, one with relative residual 9.71×10^{12}
Divided success by the whole corpus	Conjugate Gradient’s true conditional success rose from a reported 10.4% to 84.3%
Refinement given to one method only	Accuracy is now an orthogonal, universally applied factor (Section 6.5)
Re-factored inside the iteration loop	Gauss-Seidel and SOR runtime fell by a factor of 11 to 14
An overly strict divergence guard	cdde6 and similar non-normal cases (peak 44,764×) were being wrongly recorded as divergent
Two copies of the library drifting apart	Notebooks are now generated, never hand-edited, and the accidental copy removed from the repository
Downloaded files never verified against their own header	Six truncated files (Section 3.2) caught before silently corrupting a run
Trusted two scipy routines without local testing	The four hangs of Section 5.7, one costing an entire twelve-hour cloud session before being traced

Every one of these is, in miniature, the same failure: a piece of code’s own claim about itself, a return flag, a percentage, a silent success, was trusted in place of an independent measurement. The remedy applied uniformly across all eight was the same: compute the ground truth independently (the residual, the file’s own declared nonzero count, a local test on the full corpus) and never let a component certify itself.

7.5 Why the bottleneck objective wins its own metric and still barely converts systems

This is the first of the project’s audit-era findings, and it began as a puzzle rather than a planned result. Section 5.8 reported that bottleneck beats MC64 365 to 0 (522 ties) on the worst row ratio, the exact quantity it was built to minimise, yet converts only two additional systems over MC64 across the whole corpus. A result that wins its own metric this decisively and moves the actual outcome this little demanded an explanation, not a shrug.

The explanation is a threshold effect. The row-ratio bound certifies convergence only below 1, and says nothing whatsoever about behaviour above it: MC64 already reaches a worst ratio under 1 on 43 matrices, bottleneck reaches it on the identical 43, and neither objective crosses that line on a single further matrix in either direction. Every one of bottleneck’s 365 wins happens far above the threshold that would actually matter: 20 wins where MC64’s own ratio sits between 2 and 10, 175 wins between 10 and 10^3 , and 170 wins above 10^3 . The median win improves the ratio from 597 to 368, a 35.8% improvement, on matrices whose Gauss-Seidel iteration matrix has a median spectral radius of 117.8 in that same population, a value so far past 1 that no bound on the ratio could possibly certify anything useful about it.

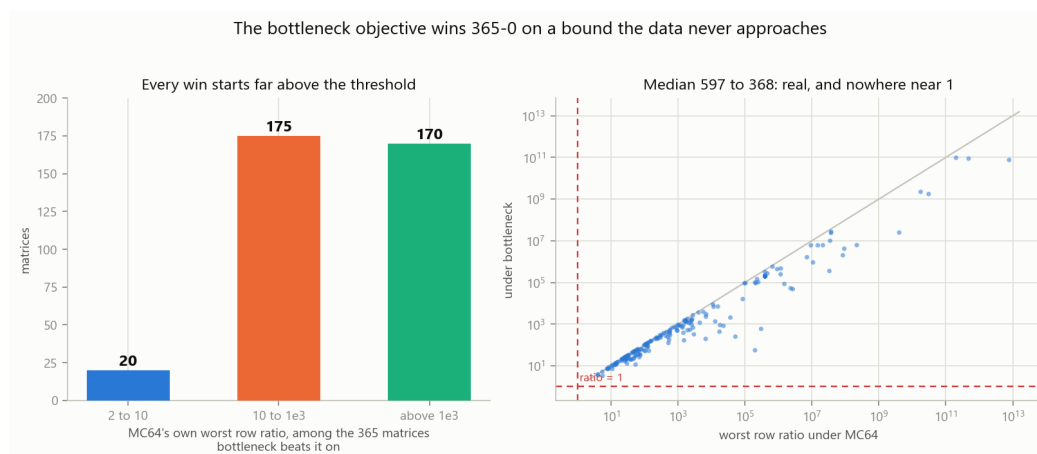


Figure 28: Left: the 365 bottleneck wins over MC64, binned by the magnitude of MC64's own worst ratio. Right: the same wins as a log-log scatter against the ratio=1 threshold that would need to be crossed for the win to matter to convergence.

A second, independent measurement sharpens exactly how much of the corpus this bound actually reaches. Of 125 matrices with $\rho(T_{GS}) < 1$, only 23 are strictly diagonally dominant, 18.4%; four convergent matrices in five converge for reasons the row-ratio bound cannot see at all. Within the strictly-dominant regime itself, the ratio predicts $\rho(T_{GS})$ almost perfectly, a Spearman correlation of +0.988 across 23 matrices; outside that regime the correlation falls to +0.576, and 42% of matrices outside the regime converge anyway. Optimising a sufficient condition is worth very little when the condition is a step function and the data sits nowhere near the step. The remedy this project tried in response, targeting the spectral radius directly instead of a bound on it, is Section 7.6. The finding itself, that bottleneck is an exact solution to a well-posed but poorly targeted problem, and that 365-0 is what being exactly right about the wrong quantity looks like, is treated in this report as a genuine negative result rather than an embarrassment, because it generalises past this specific project: any diagonal-dominance-style bound applied to a real sparse corpus is likely to meet the same wall.

7.6 Why targeting the spectral radius directly still only converts two systems

If the bound is the problem, the natural next attempt is to target what actually decides convergence, the spectral radius $\rho(T_{GS})$ itself, rather than a sufficient condition on it. $\rho(T_{GS})$ is not an assignment problem and cannot be solved exactly the way the row-ratio objective can, but it can be searched: starting from the pipeline's own chosen permutation, a pairwise row-swap descent repeatedly swaps two rows and keeps the swap only if a power-iteration estimate of $\rho(T_{GS})$ falls (`tools/probe_rho_search.py`), then every predicted improvement is checked against the real harness rather than trusted on its own (`tools/verify_rho_search.py`), a methodological discipline adopted specifically because of the harness lessons in Section 7.4.

The search works as a search: across 195 matrices it lowered the estimated $\rho(T_{GS})$ on 154 of them, by a median of 19.9%, and pushed 22 matrices from an estimated value above 1 to an estimated value below it, reaching further than an earlier back-of-envelope estimate had suggested (that estimate, that roughly 40 matrices sat within 20% of the threshold after the portfolio's own permutation, undercounted what a direct search could reach). It is also cheap, 0.73 seconds median per matrix, 195 seconds for the entire probe.

predicted crossings: 22 actually solved: 2 lost: 0

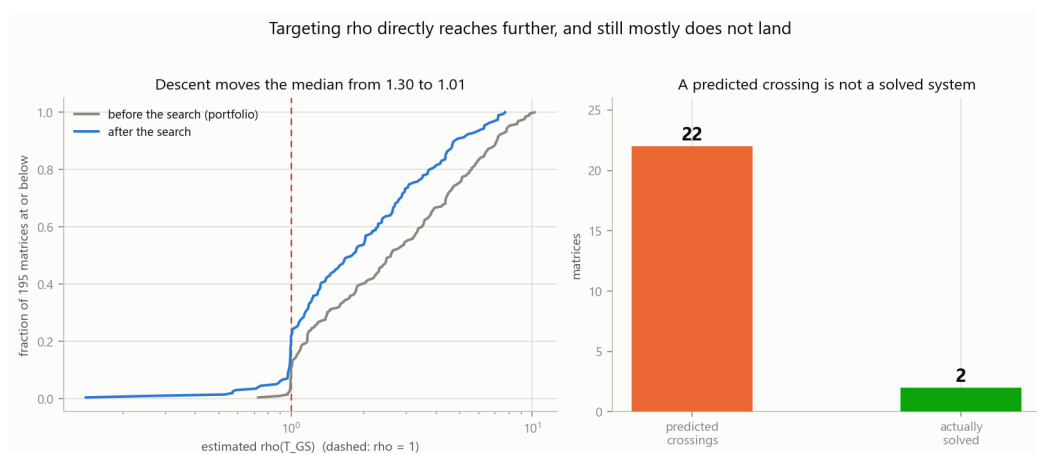


Figure 29: Left: the empirical distribution of the estimated spectral radius before and after the rho-targeting search. Right: the 22 matrices predicted to cross below $\rho = 1$ against the 2 that actually solved once checked against the real harness.

Two reasons explain the gap between 22 predicted and 2 verified, and both matter more than the headline number itself. First, a spectral radius just under 1 is not convergence within any practical iteration budget; the 22 crossings land at estimated values of 0.966, 0.976, 0.986, 0.991, and 0.9996, asymptotically guaranteed but unreachable inside a ten-thousand-iteration cap, exactly the same failure mode already identified in Section 7.2 for the classical hypothesis classes. Second, the power-iteration estimator is least trustworthy exactly where the search needs it most: sixty iterations on a strongly non-normal operator cannot resolve ρ to the third decimal place, so some of the predicted 22 crossings never actually happened once measured properly, and using a ranking device as a threshold test asked more precision of it than it was ever built to deliver. Two of the twenty-two, `adder_dcop_39` and `rand_001_n5` (a synthetic five-by-five system), did genuinely cross and solve; one solved case among a corpus of hundreds of thousands of candidate matrices is a small but real and honestly earned gain, not a rounding artefact.

7.7 Why the ILU hole claim was retracted

This is the most consequential entry in this section, because it is a retraction of a headline result the project had already reported, and it is told here in full, both what was claimed, what the audit found, and what survives.

7.7.1 What was claimed

Section 5.9 reported that 166 matrices exist on which no incomplete LU factorization can be built at all, which disables the entire preconditioned solver family at once, and that the same diagonal-selection pipeline that helps the stationary methods makes 150 of those 166 matrices constructible.

7.7.2 What the audit found

The routine that builds an ILU preconditioner, `build_ilu`, retried scipy's `spilu` factorization at three settings on failure, and every retry *increased* the drop tolerance. `spilu` fails on encountering a zero pivot, and dropping more small entries makes a zero pivot more likely to appear, not less; the retry ladder was retrying in precisely the direction that makes failure more probable. It never varied the column-permutation strategy or the diagonal pivoting threshold either, the two levers that actually matter for this kind of failure.

Re-running all 166 matrices against stock `spilu`, with no permutation applied at all, tells the real story.

Table 8: Stock spilu factorization success across the original 166 “impossible” matrices, at eight settings, with no permutation applied.

Setting	Usable factorization
ours_1 (drop 1e-3, fill 5)	0 / 166
ours_2 (drop 1e-2, fill 10)	0 / 166
ours_3 (drop 1e-2, fill 20)	0 / 166
permc_natural (drop 1e-4, fill 20)	14 / 166
full_pivot (drop 1e-3, fill 10, thresh 1.0)	9 / 166
permc_mmd (drop 1e-4, fill 20, MMD_AT_PLUS_A)	44 / 166
low_drop (drop 1e-6, fill 20)	113 / 166
no_drop (drop 0.0, fill 50)	150 / 166

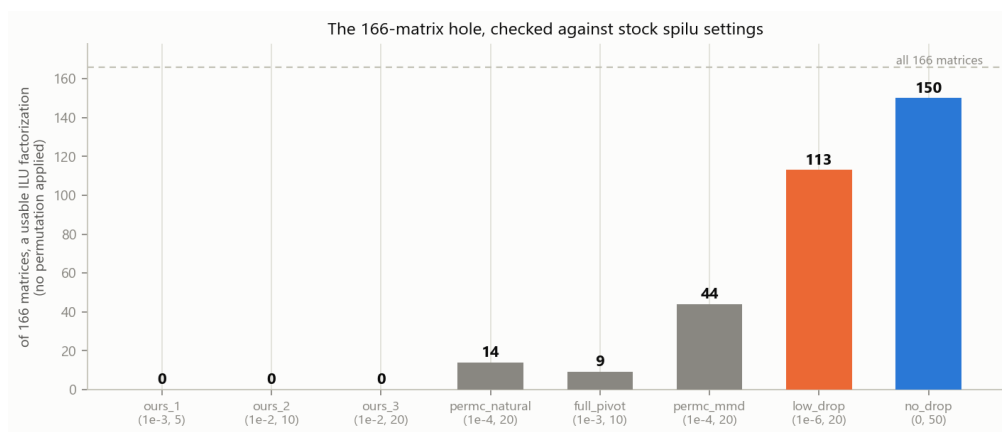


Figure 30: The eight spilu settings audited against the original 166-matrix “ILU hole,” with no row permutation applied at all. `no_drop` alone, with none of the project’s own preprocessing, factors as many of the 166 as the entire reordering pipeline was originally credited with.

One hundred fifty of 166 factor without any permutation whatsoever, using nothing more than a different, already-available scipy setting. It is, moreover, the same 150: the reordering-rescue set from Milestone 8 and the `no_drop` set from this audit differ by exactly one matrix in each direction (`eurqsa`, `west0156`), 149 in common. There is no cost defence for the original ladder either; `no_drop` takes 0.002 seconds median and 23.4 seconds in total across all 150 matrices, so it is not buying feasibility at any real price. It is simply the setting that should have been tried in the first place.

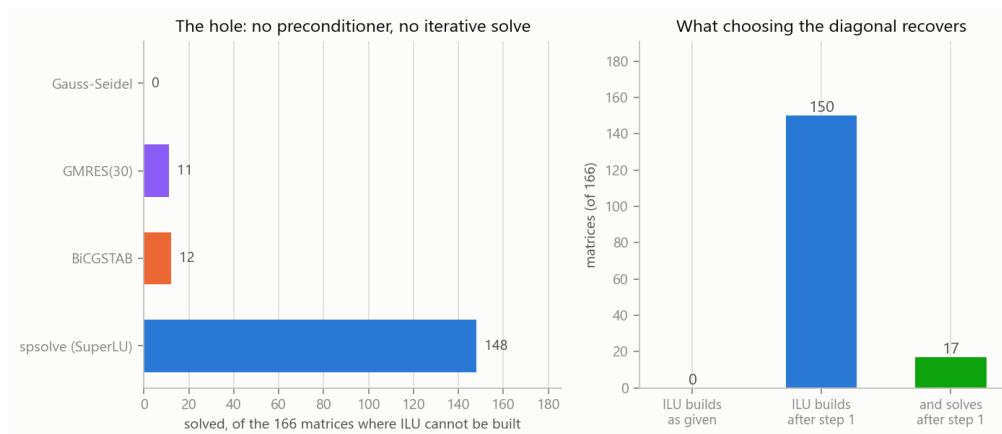


Figure 31: The original ILU-hole result as first reported. Retained here as a record of the claim that was audited, not as a validated finding; see the correction above and in Section 7.7.2.

Restricting instead to a genuinely incomplete factorization, `low_drop` at a drop tolerance of 10^{-6} rather than the near-complete `no_drop`, still factors 113 of the 166. The claim fails under either reading: whether judged against a near-complete factorization or a properly incomplete one, the fix credited to reordering was in substantial part a fix that plain, correctly configured `spilu` settings already provided for free. **Sixteen matrices resist every setting tried.** That is the real size of the hole, a tenth of what was originally reported.

7.7.3 Consequences and what has and has not been corrected

`build_ilu` has been fixed: the retry ladder now runs $(10^{-3}, 5) \rightarrow (10^{-4}, 10) \rightarrow (10^{-6}, 20) \rightarrow (0, 50)$, moving toward robustness rather than away from it, and every candidate factorization is probed on a random vector before being accepted, since a factorization that succeeds numerically and then maps a vector to `inf` was previously being returned and silently poisoning every downstream Krylov step, a real and not merely hypothetical failure mode: the `permc_mmd` setting above built 45 factorizations and only 44 turned out to be usable once probed.

What has not yet been corrected is more important to state plainly than what has. Every ILU-involving number reported anywhere in this project, the main sweep's 617, 616, and 602 solved counts in Table 3, and the full-corpus reordering-in-front-of-ILU result from Milestone 10, rests on the broken retry ladder and needs to be re-measured, not merely re-derived from existing numbers. A corrected re-run was queued on the project's cloud compute platform in three separate jobs; as of the writing of this report, all three show a cancelled-and-acknowledged status rather than a completed one, so the corrected numbers do not yet exist. This report accordingly does not state a corrected count for Result 7 of the original project record (reordering in front of ILU across the full corpus); it states only that the number originally computed there, a net gain of 18 systems with 30 gained and 12 lost, was computed under the same defect and should not be cited as final until the re-run lands. The honest, load-bearing lesson of this milestone is not the specific count of 166 versus 16; it is that the project's own diagnostic tooling attributed a fix to its novel preprocessing step when a substantial majority of that fix was actually available from a baseline library call that had simply never been tried correctly, and that this was only caught because the group chose to audit its own claim before submission rather than after.

7.8 What all three attempts add up to: applicability, not convergence

Three independent attempts to improve convergence by permuting rows are described in this report: the bottleneck objective (Section 7.5), the direct spectral-radius search (Section 7.6), and, as a point of comparison, MC64 against min-sum, which Section 5.6 already proved are the identical objective under two names. All three converge on the same conclusion.

Table 9: Three independent attempts to improve convergence by row permutation, and what each one actually converts.

Approach	What it optimises	Systems converted
Bottleneck objective	the diagonal-dominance bound, exactly (365-0 against MC64)	+2
Spectral-radius search	$\rho(T_{GS})$, by direct pairwise descent (22 crossings predicted)	+2
MC64 against min-sum	the product of the diagonal, in two disguises	0

And yet the full pipeline recovers 199 systems, not a handful. Where does that number actually come from? Decomposing every recovered (matrix, method) pair by whether the diagonal was already usable before reordering answers the question directly.

Table 10: The 199 systems the pipeline recovers, decomposed by whether reordering was fixing an empty diagonal or improving one that already existed.

Method	Gained	Diagonal empty before	Diagonal full before
Jacobi	48	44 (92%)	4 (8%)
Gauss-Seidel	86	75 (87%)	11 (13%)
SOR	65	63 (97%)	2 (3%)
Total	199	182 (91%)	17 (9%)

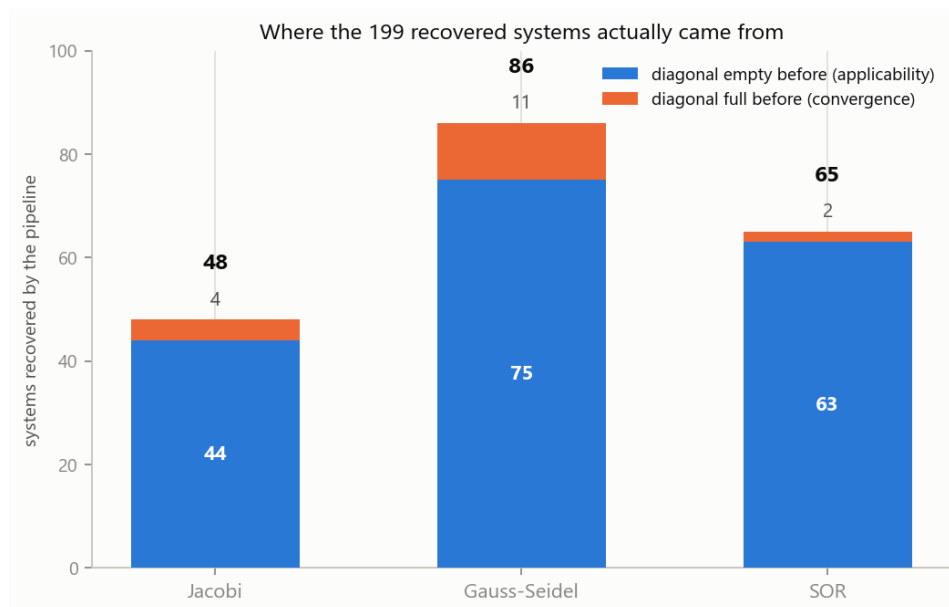


Figure 32: Every gain the pipeline reports, split into applicability (a zero diagonal made usable) against convergence (an already-usable diagonal made to converge), per method.

Ninety-one percent of the gain is applicability, a zero-free diagonal supplied where there had been none, exactly what any perfect matching provides whenever one exists. Only 17 of 199 recovered systems are cases where a usable diagonal already existed and was genuinely made to converge faster or at all. This is not a weaker result than originally advertised; it is the same result with its actual mechanism identified, and that mechanism is corroborated three separate ways, by the bottleneck-versus-MC64 comparison, by the direct spectral-radius search, and by this decomposition computed independently of either. Row permutation, on this corpus, buys applicability, essentially for free, and buys convergence only at the margins, a handful of systems at a time, because the near-1 spectral-radius wall documented in Section 7.2 sits in the way of every permutation-based remedy this project tried.

7.9 Known negative cases, reported rather than hidden

A complete failure analysis has to include the cases where the project's own remedy made things worse, not only the cases where it helped. `a_ss` (chemical process simulation, $n = 53$) is the one matrix in the corpus where the bottleneck objective increases the spectral radius rather than lowering it, from 1.884 to 29.605: protecting the single worst row in a min-max objective can come at the direct expense of every other row, and this is exactly what happens here. The portfolio's power-iteration estimate of $\rho(T_{GS})$ also misranks near-ties on occasion, costing two rescues that a more precise (and more expensive) estimate would have caught. Adaptive omega, on top of reordering, loses 12 systems relative to reordering alone for reasons already described in

Section 5.10: choosing ω per matrix has no “do nothing” fallback the way the diagonal selector does. None of these losses appear in a subset probe restricted only to matrices that could not be solved at all beforehand, which is precisely why the full 927-matrix pipeline run, rather than any smaller probe, is the number this report treats as authoritative wherever the two disagree.

8 Discussion

8.1 Strengths

The project's central empirical claim, that the base paper's conclusions do not survive contact with real engineering matrices at realistic sizes, is supported from several independent angles rather than one: the direct base-paper replication in Section 6.3, the spectral-radius validation in Section 6.4, and the theorem-coverage analysis in Section 7.2 all point at the same underlying cause, a spectral radius that clusters just above 1 for the large majority of real matrices where no classical convergence hypothesis holds. The preprocessing pipeline is a genuine, measured contribution: 199 systems recovered across a 927-matrix corpus, with zero systems lost on the diagonal-selection step, is a result that stands regardless of how the ILU correction eventually lands. Perhaps the strength we are most confident stating plainly is methodological rather than numerical: the project caught and corrected eight defects in its own measuring instrument (Section 7.4) and, in its final phase, audited and partially retracted one of its own headline claims (Section 7.7) before submission rather than after. A benchmark that survives its own authors trying to break it is worth more than one that never had anyone try.

8.2 Limitations

The most important limitation is the one already stated as plainly as we can manage in Section 7.7: every ILU-involving number in this report rests on a retry-ladder defect that was found too late in the project's timeline to re-measure before submission, and the corrected re-run had not completed as of this writing. A second, related limitation is that the corpus, though drawn from real engineering matrices rather than random ones, is a sample of 927 out of 1,184 SuiteSparse matrices meeting the size criterion, not a complete census; a further 353 matrices, mostly of graph-theoretic origin, have been downloaded but deliberately not mixed into this corpus, since doing so partway through the project would have silently invalidated every number already measured. A third limitation concerns the project's own remedy rather than the corpus: every permutation-based attempt at improving convergence, the bottleneck objective, the direct spectral-radius search, and MC64 itself, converts only a handful of systems each, because the near-1 spectral-radius regime documented in Section 7.2 appears to be a genuine property of this class of real matrix, not an artefact of any one objective's design. Finally, the power-iteration estimator used throughout the adaptive relaxation factor and the spectral-radius search is, by its own construction, a ranking device rather than a precision instrument, and Section 7.6 shows concretely what happens when it is asked to do more than rank.

8.3 Future work

Three directions follow naturally from what this project found. The corrected ILU re-measurement is the most immediate: with `build_ilu` fixed, a clean re-run across the full corpus would finally answer, honestly, how much the preconditioned family actually gains from reordering, rather than the understated figure this report is forced to withhold. A second direction is suggested directly by the near-1 wall itself: since permutation alone cannot move a matrix's spectral radius far enough to matter, a natural next lever is one this project deliberately left out of scope, a genuine change to the matrix's numerical content rather than merely its row order, for instance a mild diagonal shift or an incomplete factorization used as a stationary-method preconditioner in its own right rather than only ahead of a Krylov method. A third direction is simply scale: running the same sixteen methods and the same pipeline against the remaining 353 downloaded but unused SuiteSparse matrices would test whether the 91%-applicability, 9%-convergence decomposition of Section 7.8 is a property of this corpus in particular or of real sparse matrices in general, a distinction this project's own data cannot settle on its own.

9 Conclusion

This project set out to ask a narrow question, whether a well-derived piece of classical convergence theory for Jacobi and Gauss-Seidel holds up once it is tested against matrices that come from actual engineering problems rather than a computer's random-number generator, and it ended up answering a broader one instead. The theory holds up exactly where the classical hypotheses that justify it hold, and the trouble is that those hypotheses cover a small and shrinking fraction of what a real sparse matrix looks like. On this corpus, 53% of matrices cannot even be handed to Jacobi, Gauss-Seidel, or SOR before a single iteration, because of an arbitrary accident in how their rows happen to be ordered, and of the matrices that can be handed to them, the ones no classical theorem covers succeed only 3% of the time.

Both of those numbers turned out to be addressable, though not equally so. The first, an arbitrary and structural failure, yields almost completely to a preprocessing step that costs nothing to try and cannot make anything worse: choosing which row sits on the diagonal recovers 199 systems across the corpus, with zero lost, and 91% of that recovery is exactly this kind of fix. The second, a genuinely mathematical failure rooted in where the spectral radius of real matrices actually sits, resisted every remedy this project tried. An exactly solved bottleneck objective that wins its own target metric 365 times to zero converts only two additional systems. A direct search on the spectral radius itself, rather than a bound on it, predicts twenty-two crossings and verifies two. Both numbers land in the same place for the same underlying reason: real matrices near this threshold sit close enough to be nudged mathematically but far enough that the nudge lands inside a regime no realistic iteration budget can reach.

We do not think this is a disappointing place for the project to end up. A negative result with a clearly identified mechanism, corroborated three separate ways, is worth more than a positive result whose cause is left unexamined, and the retraction described in Section 7.7 exists in this report specifically because we would rather submit a smaller claim we can stand behind than a larger one we cannot. What this project leaves behind, beyond the specific numbers, is a benchmark harness built to be honest about its own limits, a preprocessing pipeline that measurably helps and says plainly where it does not, and a documented, receipts-backed account of exactly how each of those things was found, including the parts that did not survive scrutiny. That, more than any single percentage in the tables above, is what we would want a reader of this report to take away from it.

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